PRINCIPLES OF DRUG ACTION

Targets for drug action
Quantitative drug receptor interactions

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Objectives

At the end of the lecture, the student should be able to

- Discuss interaction of drugs with their receptors
- Discuss 'message propagation' following ligand binding- Signal transduction pathways
- Discuss quantitative relations between dose or concentration of drug and pharmacologic effects

REFERENCES AND FURTHER READING

- Goodman & Gilman's Pharmacology – 12th Edition
- Katzung, Basic & Clinical Pharmacology – 14th Edition
- Rang & Dale's Pharmacology - 8TH Edition
- Color Atlas of Pharmacology 2nd Edition
- Principles of Pharmacology - The Pathophysiologic Basis of Drug Therapy-Lippincott Williams & Wilkins (2011)

- Goodman & Gilman's Pharmacology – 12^{th} Edition
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Terminologies

- Pharmakon-Drug or poison
- **Drug**: Any molecule, used to alter body's physiology and function.
  - Used to *diagnose*, *cure, treat or prevent disease* (therapeutics)

Nature & sources of drugs

Mineral	Liquid paraffin, magnesium sulfate, etc
Animal	Insulin, Thyroid, etc.
Plant	Morphine, Quinine etc
Synthetic	Aspirin, Sulfonamides, etc.
Micro-organisms	Penicillin & other antibiotics.
Drugs produced by genetic engineering	Human insulin, human growth, hormone etc.

Terminologies

Indications

- Conditions that enable the appropriate administration of the drug (as approved by the FDA).

Contraindications

- Conditions that make it inappropriate to give the drug
  - .....means a predictable harmful event will occur if the drug is given in this situation

Dosage

- The amount of the drug that should be given

- **Routes of Administration**
  – How the drug is given.

- **Side Effects/Adverse Reactions**
  – The drug's untoward or undesired effect

- **Mechanism of Action**
  – The way in which a drug causes its effects; its pharmacodynamics

Medical pharmacology is studying processes by which:

- a drug travels from the point of delivery to its site of action and out of the body-
  - **Pharmacokinetics**
- drugs alter body functions-
  - **Pharmacodynamics**
- Pharmacology deals with the ***fate*** and ***actions*** of drugs in the body

Toxicology - The pharmacology of poisons and the harmful effects of any drug.

The processes occurring between administration of a drug and the production of its effects may be divided into the following 3 phases.

Dose of formulated Drug

↓ ***Administration***

Disintegration of dosage form

Dissolution of drug

Pharmaceutical Phase

↓ ***Drug available for absorption (Pharmaceutical availability)***

Absorption
Distribution
Metabolism
Elimination

Pharmacokinetic phase

↓ ***Drug available for action (Pharmacological availability)***

Drug – receptor interaction

Pharmacodynamic phase

↓

Effect

DRUG NOMENCLATURE

- **Chemical name:**
  - Any typical organic chemical name.
- **Generic name:**
  - The assigned name of a drug, by which it will be known throughout the world no matter how many different companies manufacture it. This name is usually assigned and agreed upon by the **World Health Organization**.
- **Official name:**
  - This is the name by which a drug is listed in one of the following **official publications**: a) U.S. Pharmacopoeia, b) National Formulary
    - The official name is often the same as the generic name

- **Trade name:**
  - The name that is given by a particular company that is manufacturing the drug.
  - The company is the legal owner of the name.
  - Sometimes a single generic drug may be sold under 5 or 10 different trade names.

Names of Drugs

Chemical Name	**7-chloro-1, 3-dihydro-1, methyl-5-phenyl-2h-1**
Generic Name	**Diazepam**
Official Name	**Diazepam, USP**
Brand Name	**Valium®**

Sources of Drug Information

- United States Pharmacopeia (USP)
- British *Pharmacopeia (*BP*)*
- British *Pharmacopeia* Codex (BPC)
- Physician's desk reference (PDR)
- The *National Formulary* **(NF**)
  - Ghana *National Formulary* **(GNF)**
- Monthly prescribing reference

All substances are poisons; there is none that is not a poison. The right dose differentiates a poison from a remedy.

Paracelsus,
1493-1541

‘Targets for Drug Action’

'Corpora non agunt nisi fixata'

Paul Erlich

"A drug will not work unless it is bound"

Exceptions (e.g. osmotic diuretics, osmotic purgatives, antacids and heavy metal chelating agents)

Major targets for Drug Action

The primary targets for drug action are:

A. Pharmacological Receptors, including ligand-gated ion channels

B. Ion channels (Voltage-gated)

C. Enzymes

D. Transporters /carrier molecules

Brief introduction to the major types of drug targets

Brief Intro' - receptors

Receptors(Pharmacological):

- The term receptor is most often used to describe target molecules through which soluble physiological mediators such as hormones, neurotransmitters, inflammatory mediators, (cytokines etc.) produce their effects.

- Receptors are distinct from the other drug targets, because they distinctly interact with drugs that activate the receptor (agonists) and those that block activation (antagonists)

Brief Intro' - receptors

There are about 4 main types of pharmacological receptors

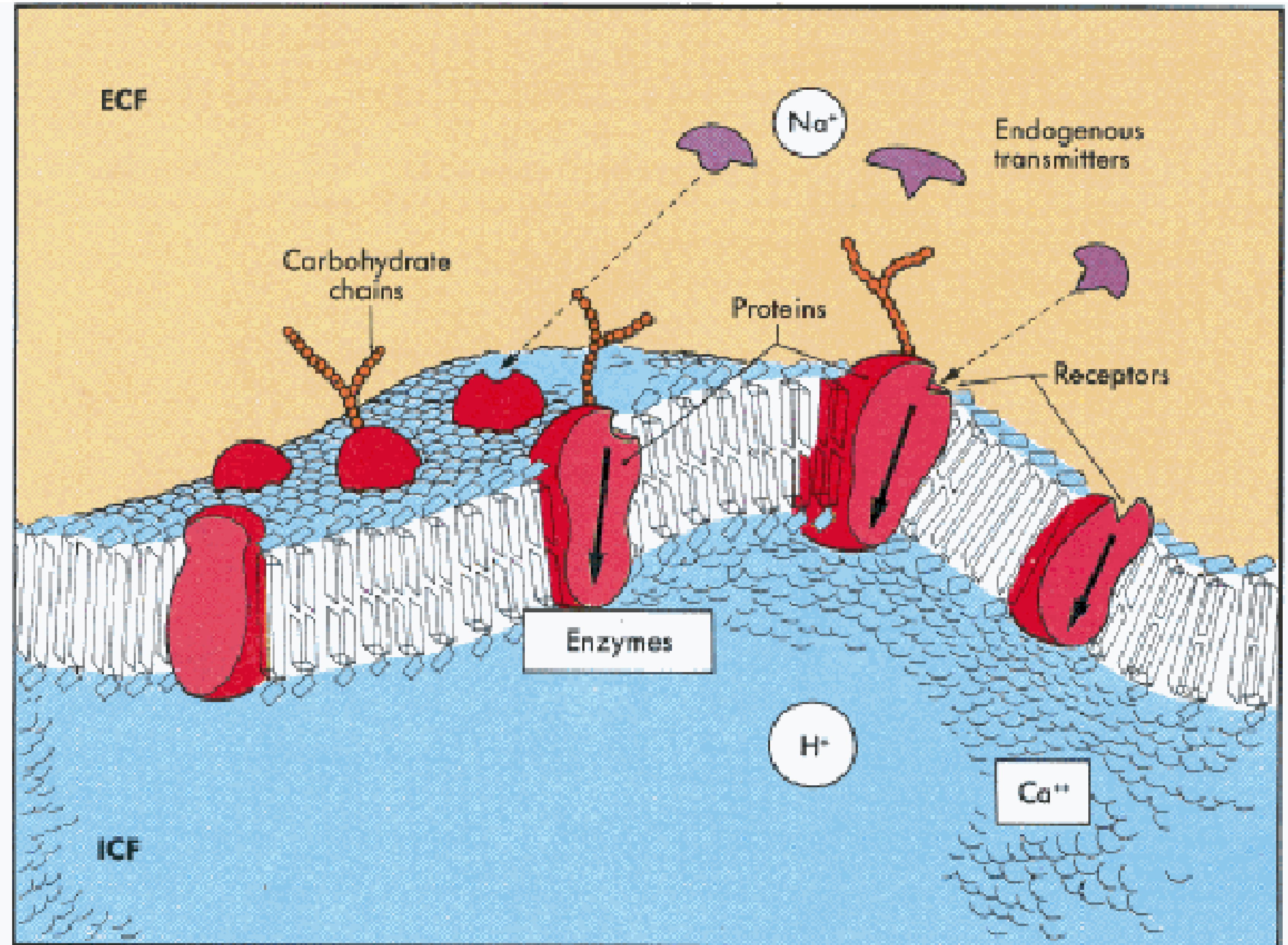
1. **G-protein-coupled receptors** – Adrenaline receptor

2. **Ion channels (Ligand-gated)**; also termed ionotropic receptors – $GABA_A$ receptor

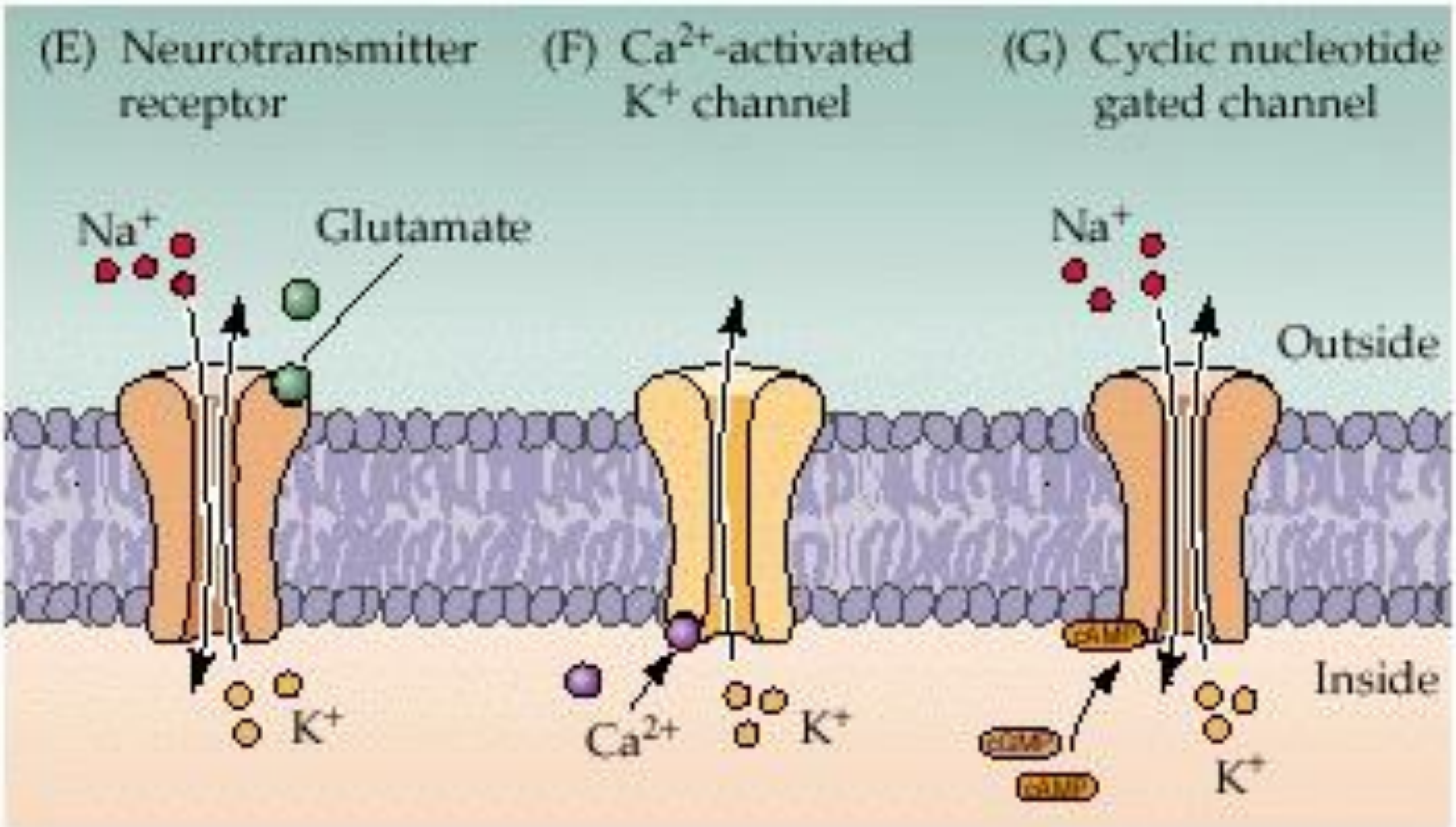
3. **Intracellular/Nuclear Receptors** – Testosterone receptor, Oestrogen receptor

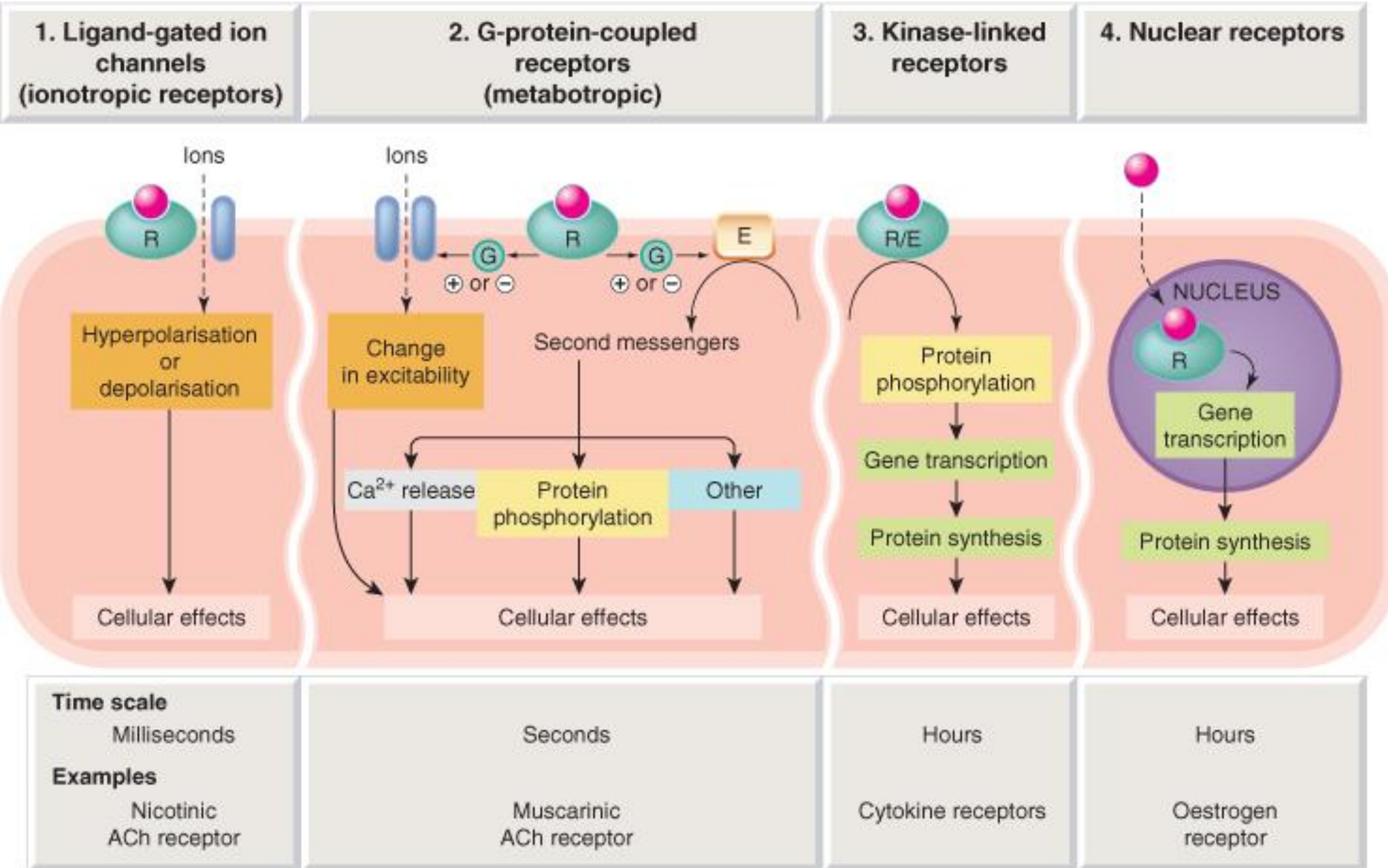
4. **Kinase-linked receptors:** these mostly initiate a kinase cascade within the cell.

Membrane-Bound Receptors

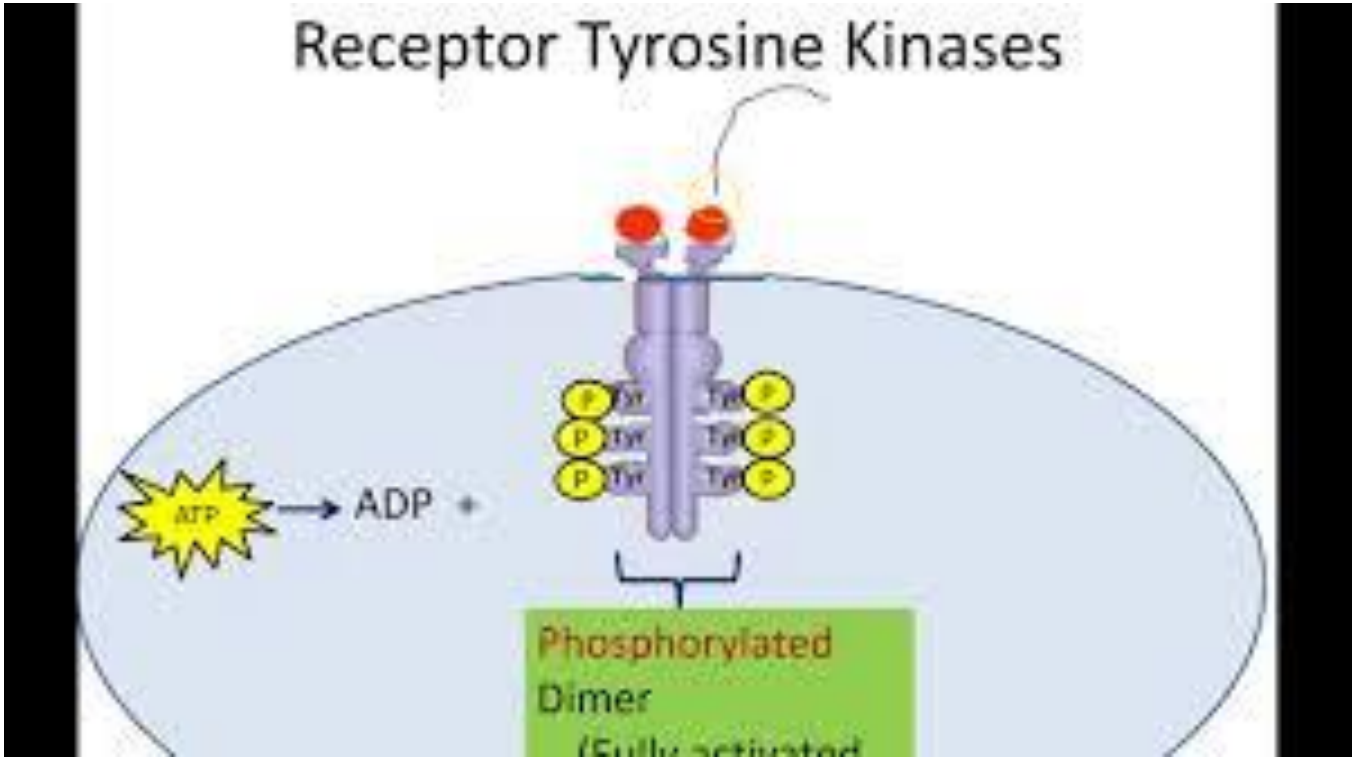


LIGAND-GATED CHANNELS

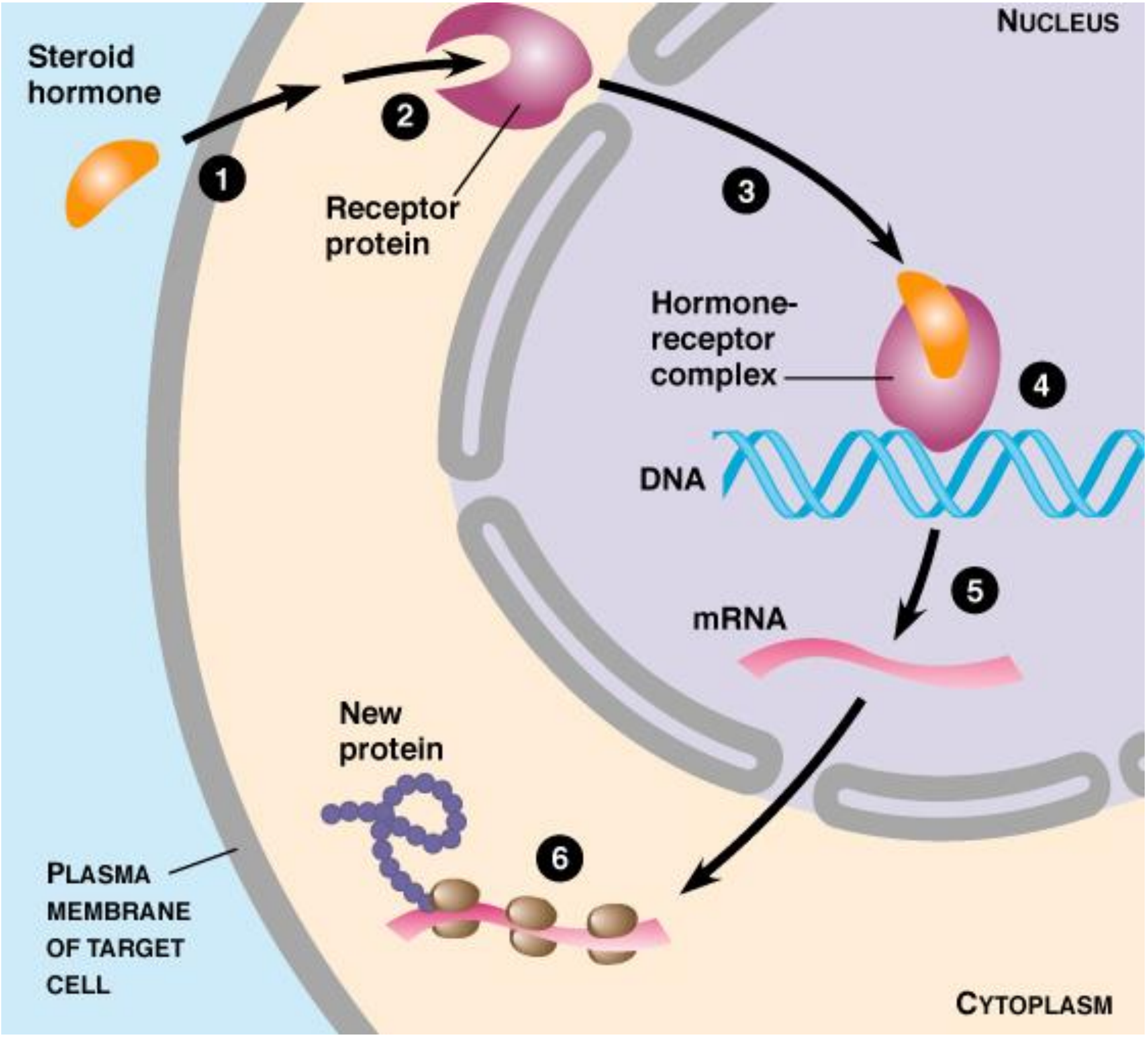



1. Ligand-gated ion channels (ionotropic receptors)
2. G-protein-coupled receptors (metabotropic)
3. Kinase-linked receptors
4. Nuclear receptors
Ions
R
Hyperpolarisation or depolarisation
Cellular effects
Ions
G
⊕ or ⊖
R
G
⊕ or ⊖
E
Change in excitability
Second messengers
Ca^{2+} release
Protein phosphorylation
Other
Cellular effects
R/E
Protein phosphorylation
Gene transcription
Protein synthesis
Cellular effects
NUCLEUS
R
Gene transcription
Protein synthesis
Cellular effects
Time scale
Milliseconds
Seconds
Hours
Hours
Examples
Nicotinic ACh receptor
Muscarinic ACh receptor
Cytokine receptors
Oestrogen receptor

Kinase-Linked Receptor



Nuclear Receptor

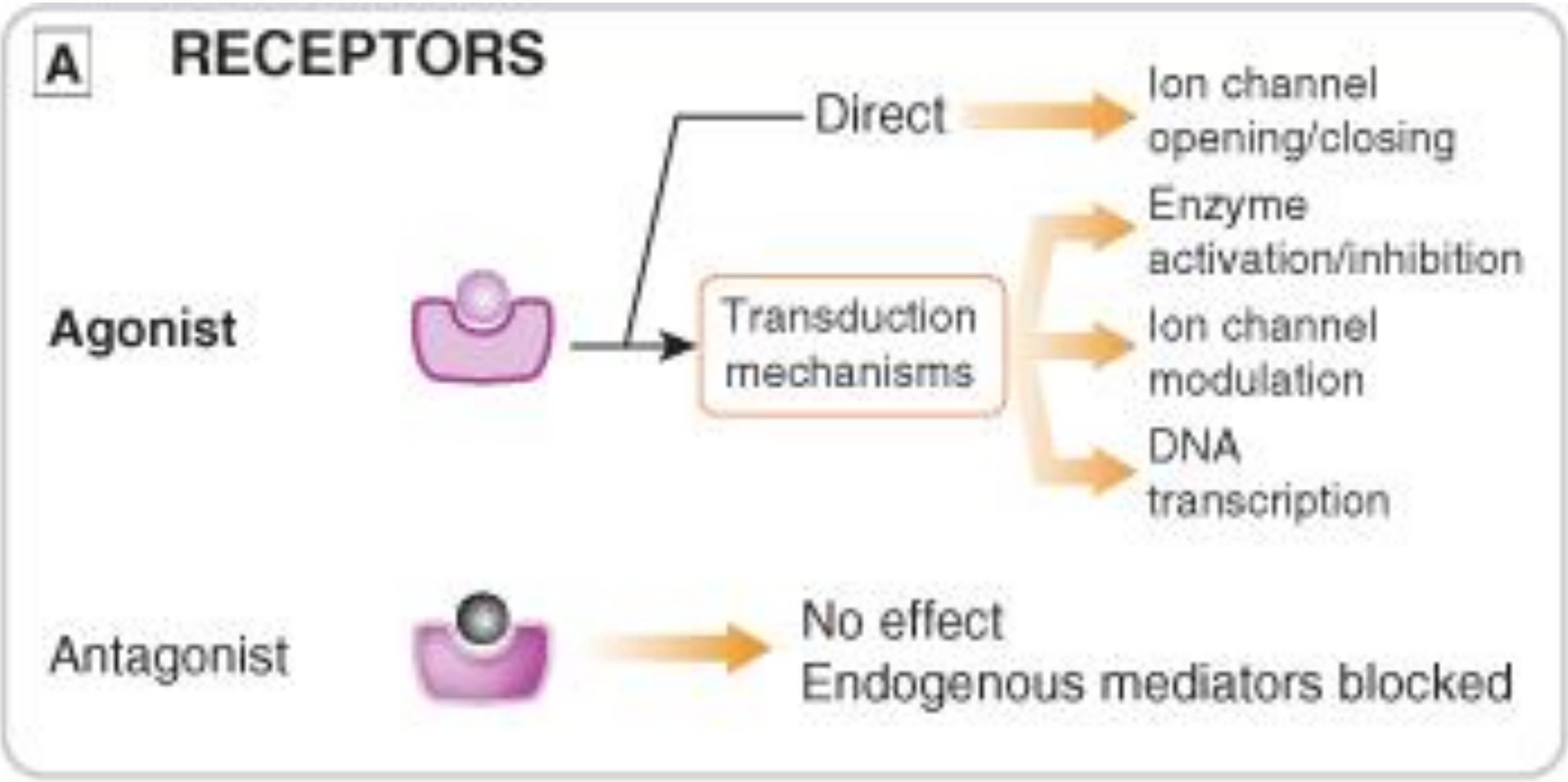


Brief Intro' - receptors

Please note:

The term **receptor** may be loosely used for all drug targets and cell surface proteins involved in communication and other functions.

Brief Intro' - receptors



Ion Channels

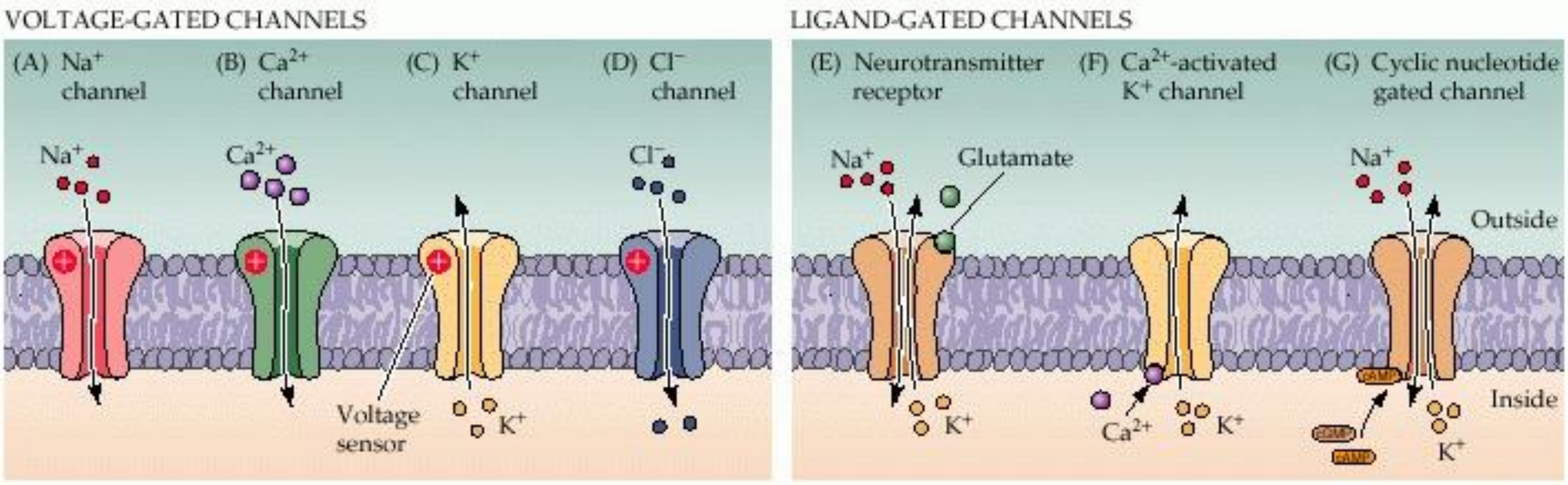
Brief Intro' - ion channels

- Ion channels are essentially gateways in cell membranes that selectively allow the passage of particular ions
- They may be induced to open when one or more agonist molecules are bound [ligand-gated (receptor-type)]

OR

- Are gated by changes in the transmembrane potential (voltage-gated).

Brief Intro' - ion channels



Brief Intro' - ion channels

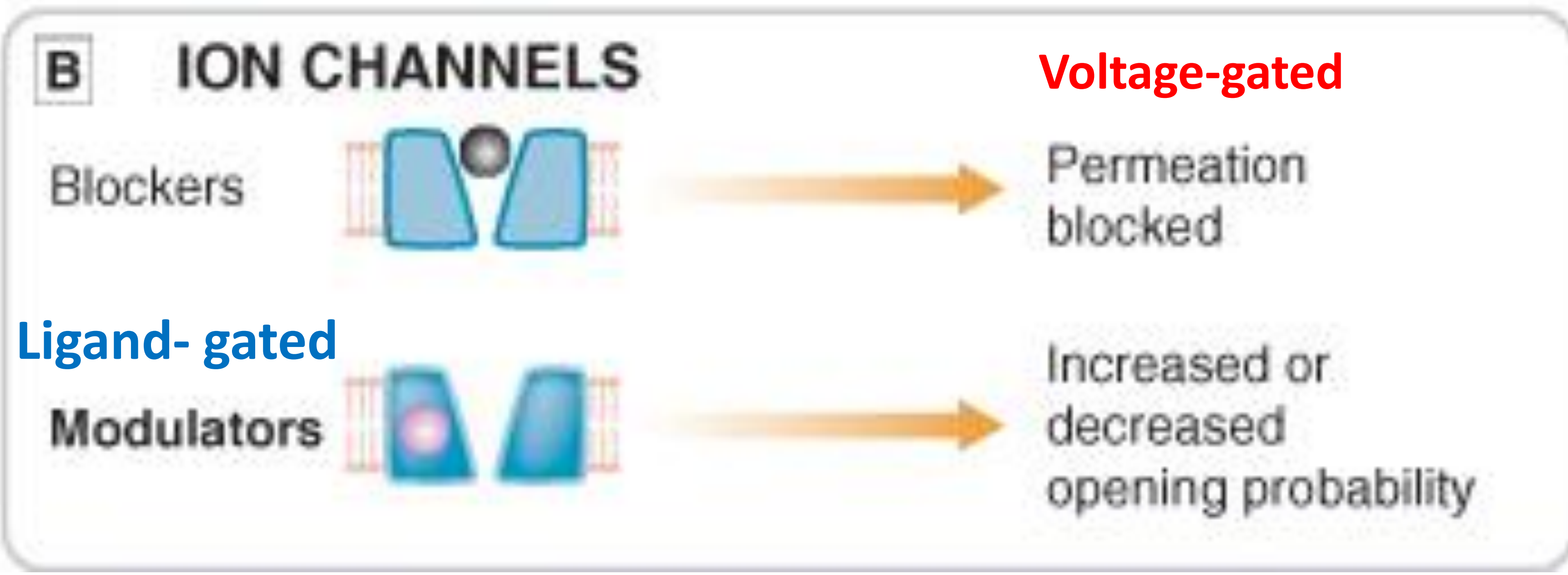
In general, drugs can affect ion channel function in several ways:

- The drug molecule may plug the channel physically, blocking ion permeation.
  - the action of local anaesthetics (procaine, lignocaine, bupivacaine ect.) on the voltage-gated sodium channel

- <u>The drug binds to the channel protein itself</u>

  - Either to the ligand-binding (*orthosteric*) site or to other sites (*allosteric*)

Brief Intro' - ion channels



Brief Intro' - ion channels

Some Drugs that bind to the channel protein itself

- **Benzodiazepine tranquillisers** (Diazepam ect.)
  - These drugs bind to a region of the $GABA_A$ receptor–chloride channel complex (a ligand-gated channel) that is distinct from the GABA binding site and facilitate the opening of the channel by the inhibitory neurotransmitter GABA.

Brief Intro' - ion channels

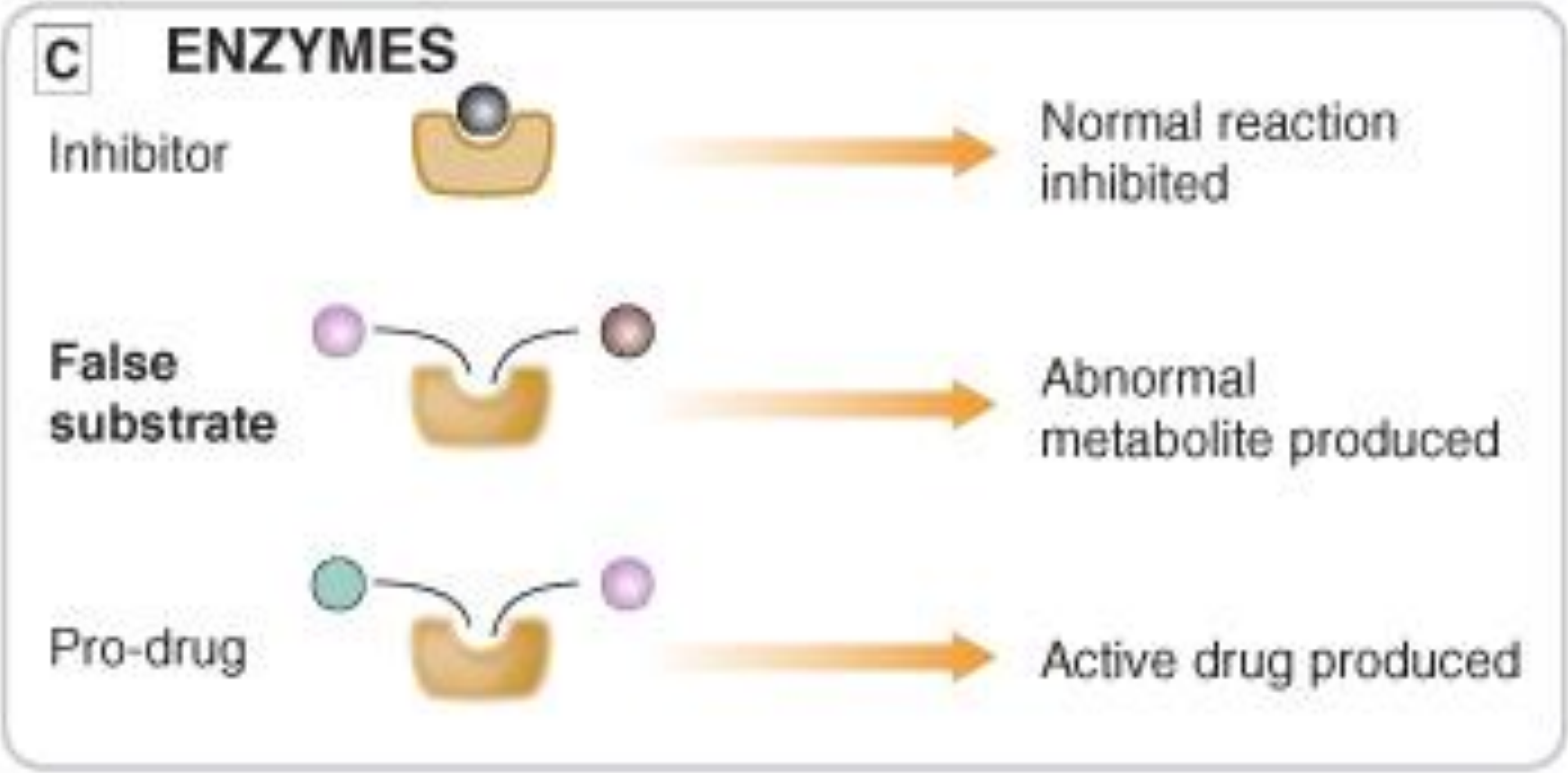
Some Drugs that bind to the channel protein itself

- **Vasodilator drugs** of the dihydropyridine type (nifedipine, amplodipine, felodipine) which inhibit the opening of L-type calcium channels

- **Sulfonylureas** used in treating diabetes, which act on ATP-gated potassium channels of pancreatic β-cells and thereby enhance insulin secretion. Gilbenclamide (Daonil), Tolbutamide, Gilclazide (Diamicron), Glipizide (Minodiab), Glimepiride (Amaryl)

Enzymes

Brief Intro' - Enzymes



Brief Intro' - Enzymes

- Many drugs act on intracellular or extracellular enzymes that synthesise or degrade endogenous ligands as

A. Inhibitor

  - *The drug molecule can be a substrate analogue that acts as a competitive inhibitor of the enzyme e.g.* ***captopril****, acting on angiotensin-converting enzyme; or irreversible and non-competitive inhibitor e.g.* ***aspirin****, acting on cyclo-oxygenase).*

Brief Intro' - Enzymes

B. False substrate

- Drugs may also act as false substrate eg. the anticancer drug **fluorouracil**, which replaces uracil as an intermediate in purine biosynthesis but cannot be converted into thymidylate, thus blocking DNA synthesis and preventing cell division

C. Activate Prodrug

- Enzyme may be used to convert prodrug to active drug e.g. **enalapril** is converted by esterases to enalaprilat, which inhibits angiotensin converting enzyme)

Table 1.4 Examples of enzymes as drug targets

Enzyme	Drug class or use	Examples
Acetylcholinesterase (AChE)	AChE inhibitors (Chapter 27)	Neostigmine, edrophonium, organophosphates
Angiotensin-converting enzyme (ACE)	ACE inhibitors (Chapter 6)	Captopril, perindopril, ramipril
Antithrombin (AT)III	Heparin anticoagulants (enhancers of antithrombin III) (Chapter 10)	Enoxaparin, dalteparin
Carbonic anhydrase	Carbonic anhydrase inhibitors (Chapters 14, 50)	Acetazolamide
Cyclo-oxygenase (COX)-1	Nonsteroidal anti-inflammatory drugs (NSAIDs) (Chapter 29)	Ibuprofen, indometacin, naproxen
Cyclo-oxygenase (COX)-2	Selective COX-2 inhibitors (Chapter 29)	Celecoxib, etoricoxib
Dihydrofolate reductase	Folate antagonists (Chapters 51, 52)	Trimethoprim, methotrexate
DOPA decarboxylase	Peripheral decarboxylase inhibitors (PDIs) (Chapter 24)	Carbidopa, benserazide
Coagulation factor Xa	Direct inhibitors of Factor Xa (Chapter 11)	Rivaroxaban
HMG-CoA reductase	Statins (HMG-CoA reductase inhibitors) (Chapter 48)	Atorvastatin, simvastatin
Monoamine oxidases (MAOs) A and B	MAO-A and MAO-B inhibitors (Chapters 22, 24)	Moclobemide, selegiline
Phosphodiesterase (PDE) isoenzymes	PDE inhibitors (Chapters 12, 16)	Theophylline, sildenafil (see Table 1.1)
Reverse transcriptase (RT)	Nucleos(t)ide and nonnucleoside RT inhibitors (Chapter 51)	Zidovudine, efavirenz
Ribonucleotide reductase	Ribonucleotide reductase inhibitor (Chapter 52)	Hydroxycarbamide (hydroxyurea)
Thrombin	Direct thrombin inhibitors (Chapter 11)	Dabigatran
Viral proteases	HIV/hepatitis protease inhibitors (Chapter 51)	Saquinavir, boceprevir
Vitamin K epoxide reductase	Coumarin anticoagulants (Chapter 11)	Warfarin
Xanthine oxidase	Xanthine oxidase inhibitors (Chapter 31)	Allopurinol

Transporters/carriers

Brief Intro' – transporter/carriers

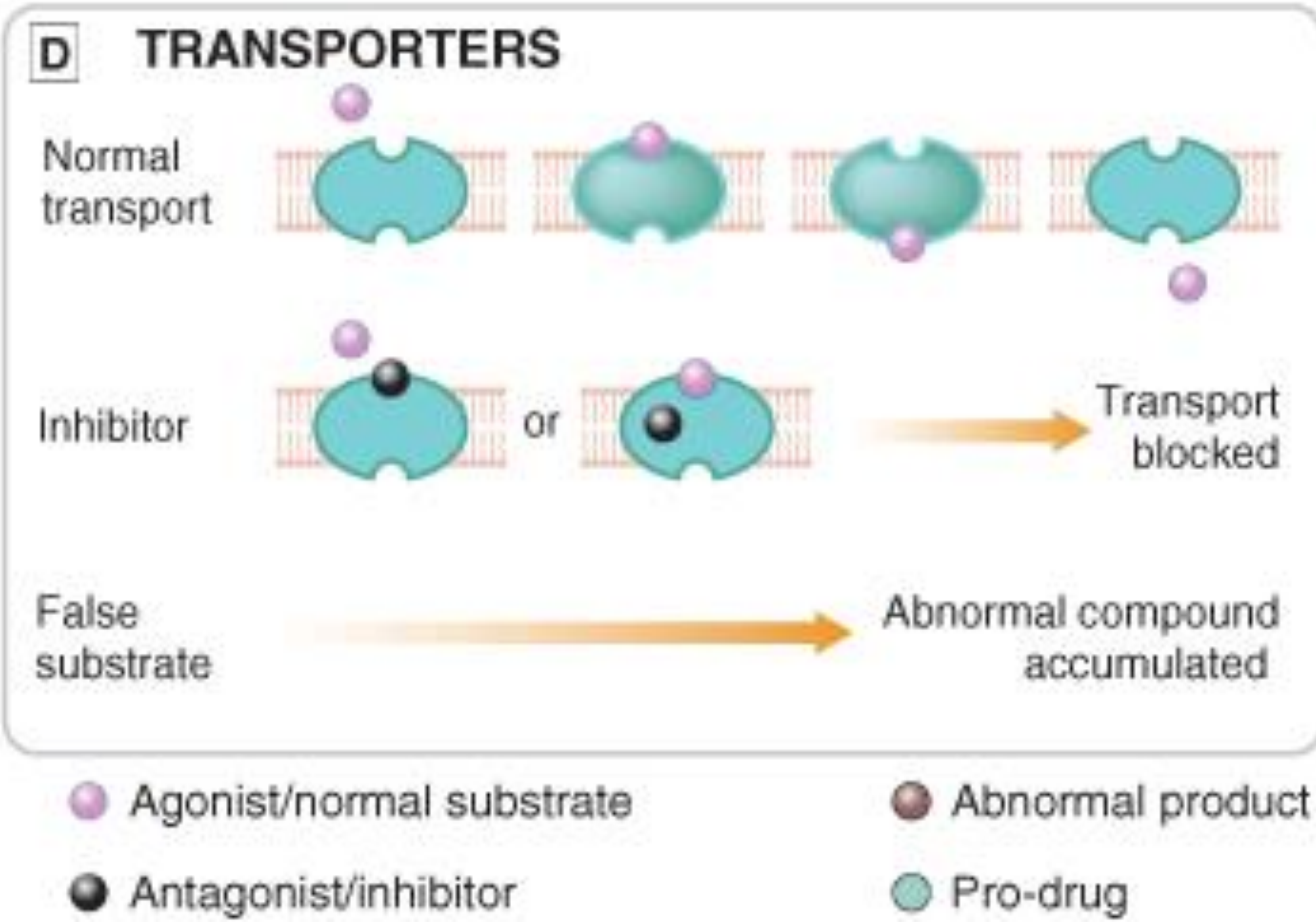
Transporter/carrier proteins

- Drugs can variously interact with carrier proteins, which transport a specific ion or organic molecule across a membrane

Brief Intro' – transporter/carriers

The carrier proteins embody a recognition site that makes them specific for a particular permeating species, and these recognition sites can also be targets for drugs whose effect is to block the transport system.

Brief Intro' – transporters/carriers



Brief Intro' - transporters/carriers

Examples include:

- Na+/Cl- co-transport in the renal tubule, which is blocked by **thiazide diuretics**

- The reuptake of neurotransmitters into nerve terminals by a number of transporters selectively blocked by classes of antidepressant drugs
    - **Bupropion** (BUP) is a dopamine (DA) and norepinephrine (NE) re-uptake inhibitor

Non-Receptor Mediated Action of Drugs

Non-Receptor Actions

- **Structural proteins** eg tubulin, actin
- **Nucleic acids**
  - DNA
  - RNA)
- **Membranes**
- **Fluid compartments**
- **pH changes**

Non - Receptor Action

- **Structural proteins** eg. Cytoskeletal proteins that are involved in cell motility (tubulin, actin)
  - **Colchicine** modulates multiple pro- and antiinflammatory pathways associated with gouty arthritis. Colchicine prevents microtubule assembly and thereby disrupts inflammasome activation, microtubule-based inflammatory cell chemotaxis, generation of leukotrienes and cytokines, and phagocytosis)

Non - Receptor Action

- **Nucleic acids**
  - **DNA**: This is very common with cytotoxic drugs used in cancer therapy, e.g. alkylating eg. Nitrogen mustards (cyclophosphamide) Nitrosoureas (streptozocin)
  - **RNA**: Currently, one of the most important RNA drugs in clinical practice are vaccines (SARS – COV2 Vaccines)

Non - Receptor Action

- **Membranes**: Cell membranes may be considered the targets for inhalation anaesthetics (e.g. diethyl ether, chloroform)

  - There is a remarkably close correlation between the ability of these agents to partition into lipid membranes and their narcotic activity. But accumulating evidence now supports direct interaction with several ion channels.

Non - Receptor Action

Fluid compartments:

- Plasma expanders such as dextran and hydroxyethyl-starch are metabolically inert polysaccharides that are osmotically active solutes used to increase blood volume

Non- Receptor Action – F' compartment

- Osmotically acting diuretics (eg. mannitol) increase urine volume and accelerate elimination of poison ('forced diuresis').
  - This sugar is quite similar to glucose in structure but does not get metabolized nor reabsorbed from the primary glomerular filtrate in the kidneys.

Non - Receptor Action

pH Changes

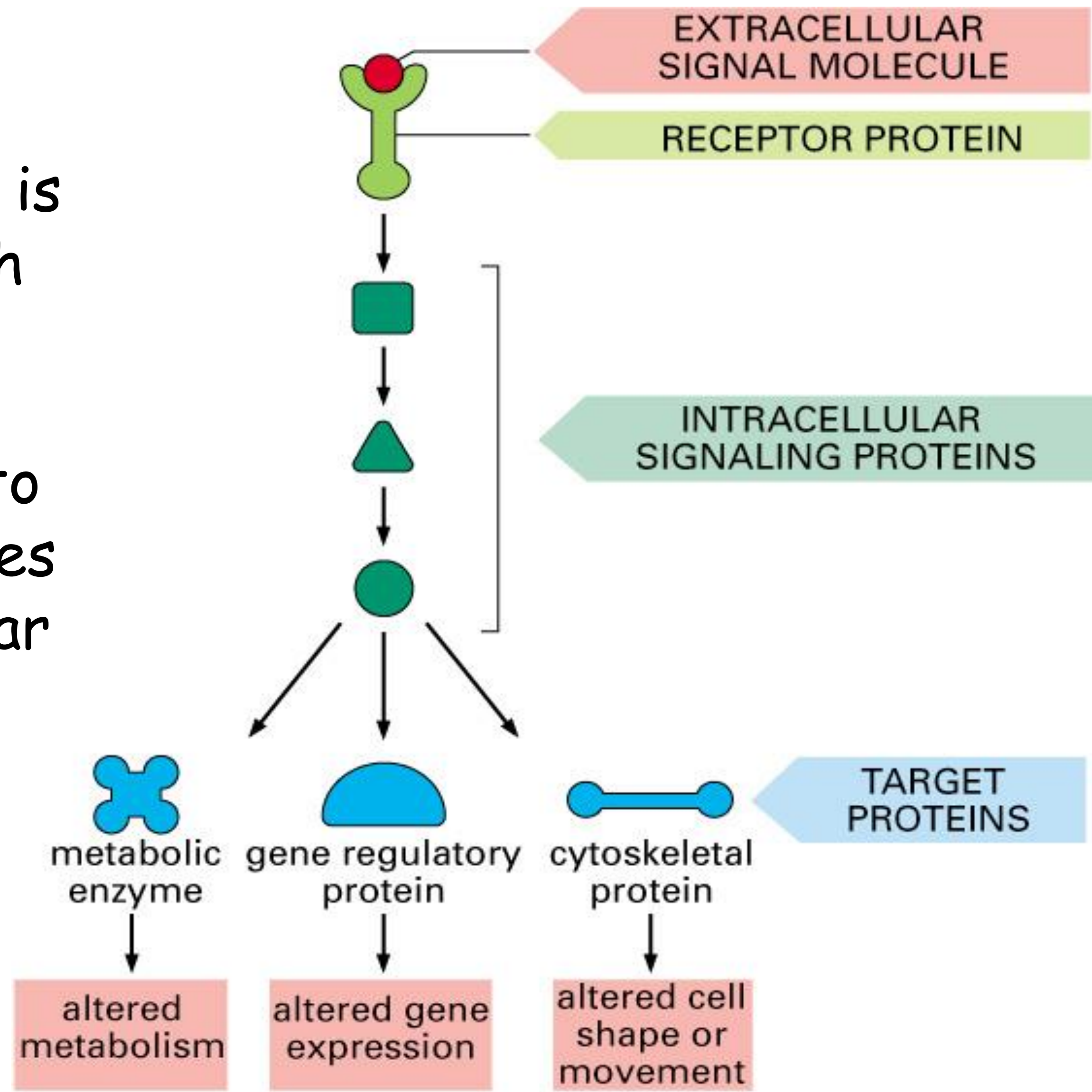
- **<u>Antiacids</u>**: Such as Aluminum Hydroxide and Magnisium Trisilicate used GIT acid reflux only neutralises protons.

'How Drugs Act'
Signal Transduction Mechanisms

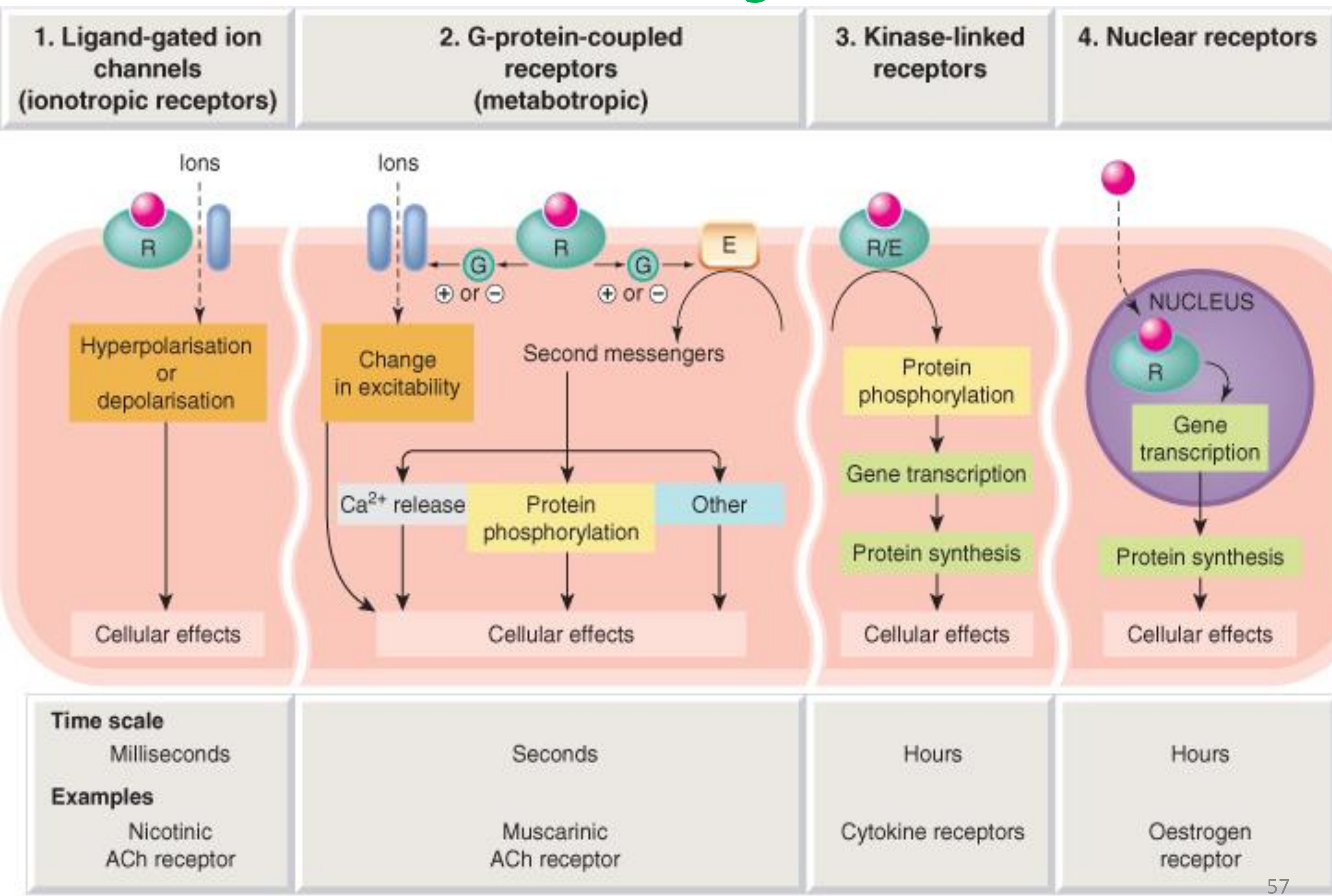
* Emphasis will be placed on **pharmacological receptors**

Signal transduction

- **Signal transduction** is the process by which extracellular inputs/signal (ligand/drug) leads to intracellular messages that modulate cellular physiology.

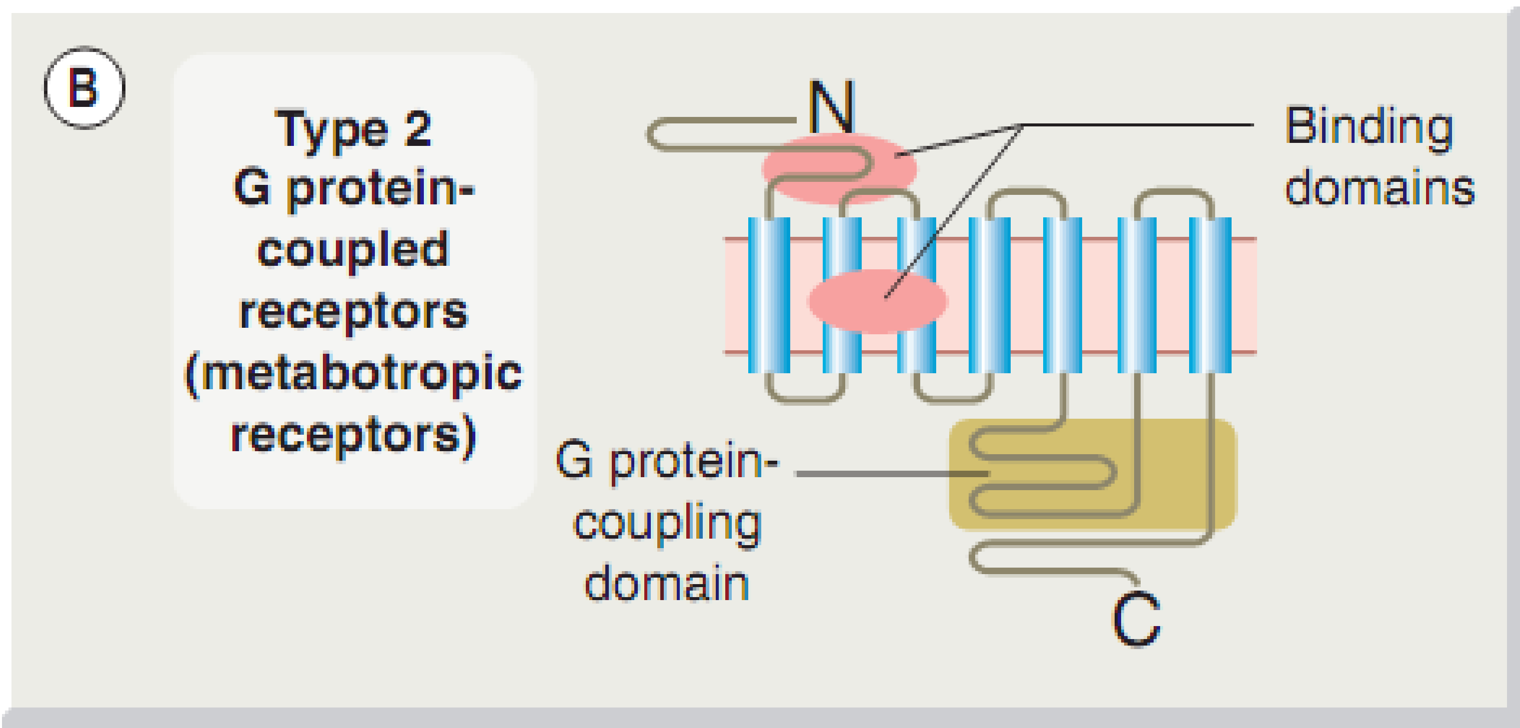


How Drugs Act

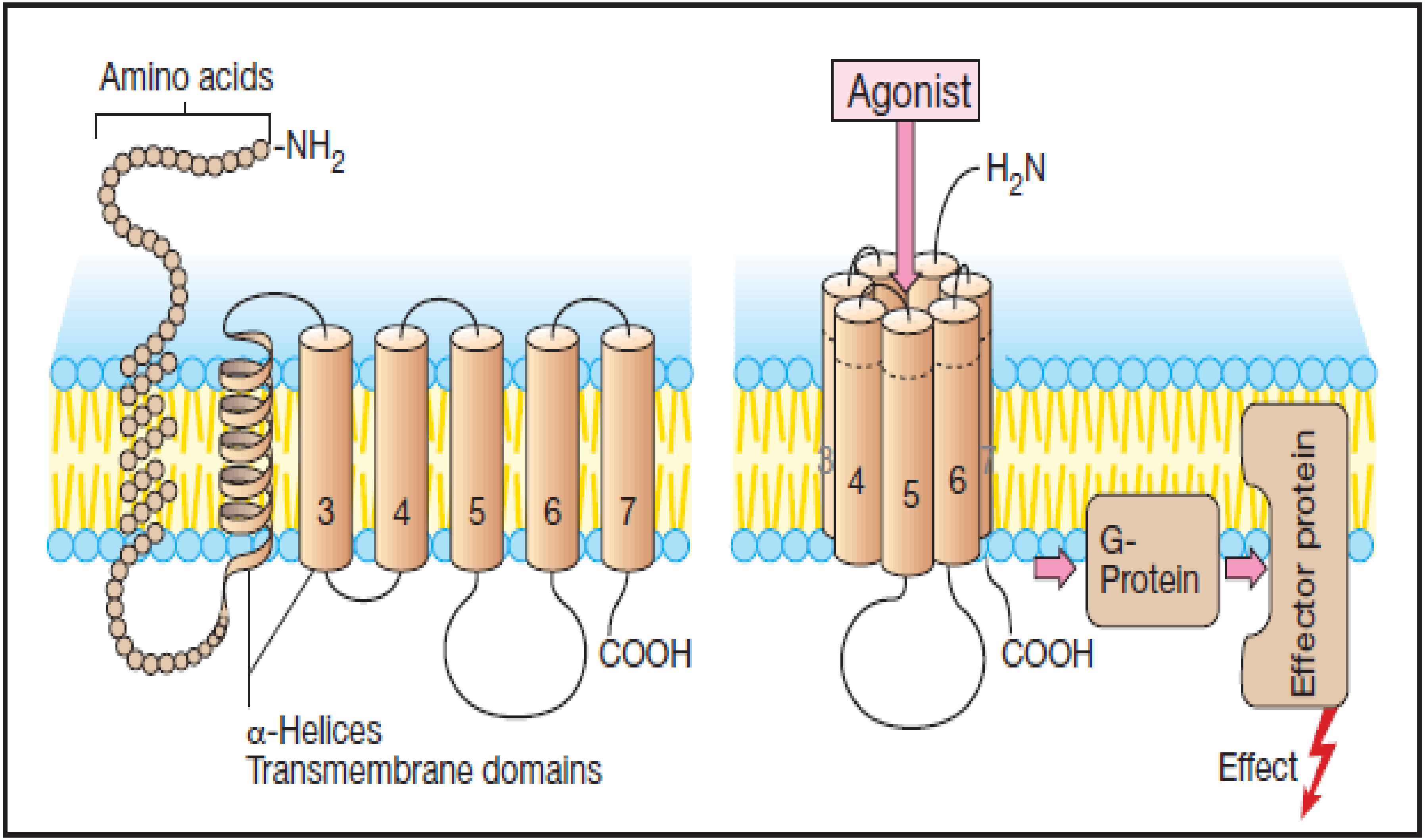


Signal Transduction : GPCRs

- GPCRs form a family of receptors with seven-transmembrane (7TM)/ seven membrane-spanning helices
  - muscarinic AChRs, adrenoceptors, dopamine receptors, 5-HT receptors, opioid receptors, receptors for many peptides purine receptors



Signal Transduction : GPCRs



Signal Transduction: GPCRs

- When a GPCR is activated, the intracellular domain interacts with guanine nucleotide-binding proteins **(G-proteins)** which modulate **second messenger molecules** linked to physiological responses.

What are G proteins?

- Proteins that bind guanosine diphosphate (GDP) and guanosine triphosphate (GTP) in its inactive and active states respectively.
- Have GTPase activity terminates its own activity.
- G-proteins modulate enzyme activity or ion channel function

Signal Transduction: GPCRs : G proteins

Subunits

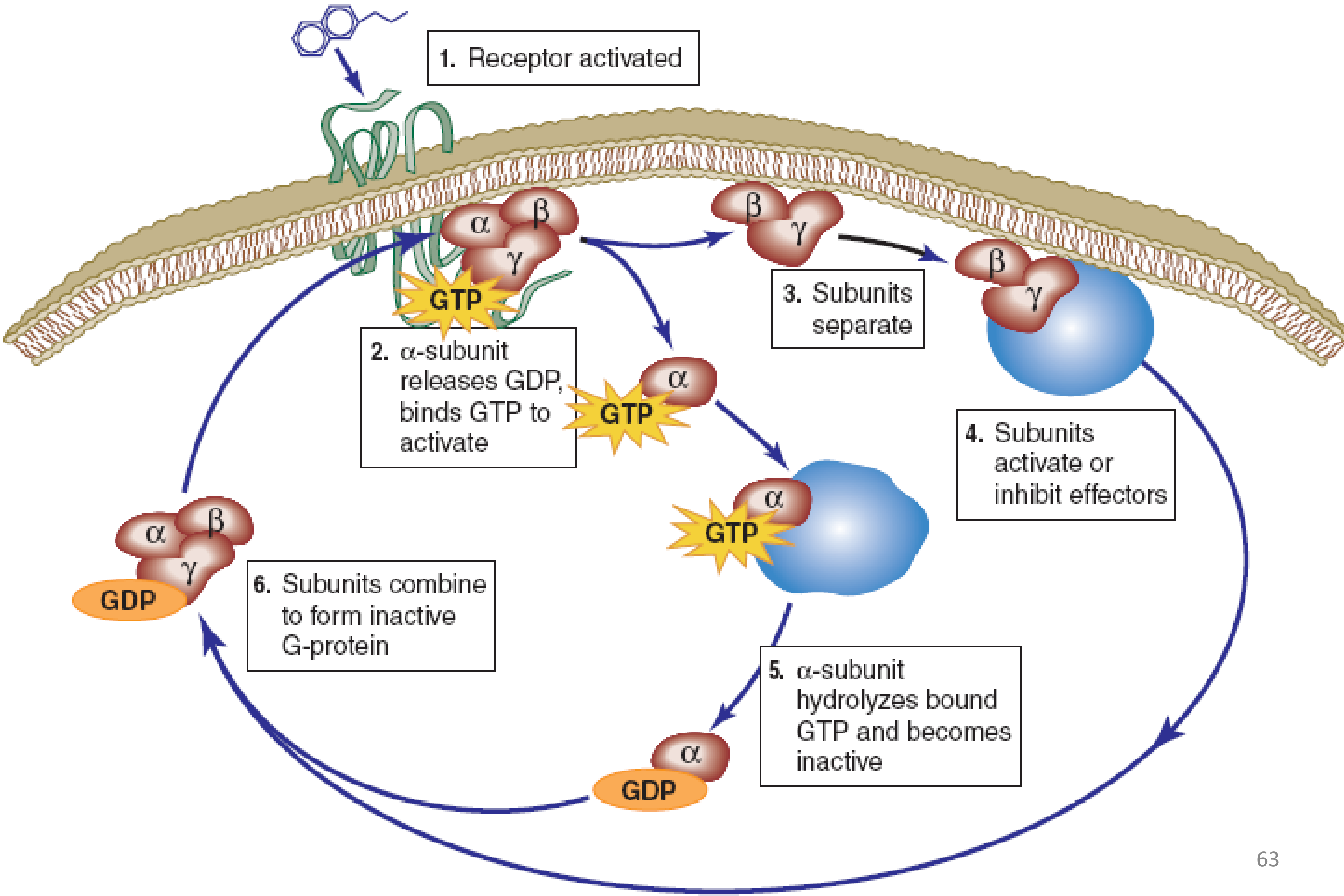
- G-protein system consists of 3 **protein subunits α** – subunit, **β** – subunit, **γ** – subunit.

- The **β** and **γ** subunits remain together as a **βγ** complex attached to the **α** – subunit when the receptor is in the resting state.

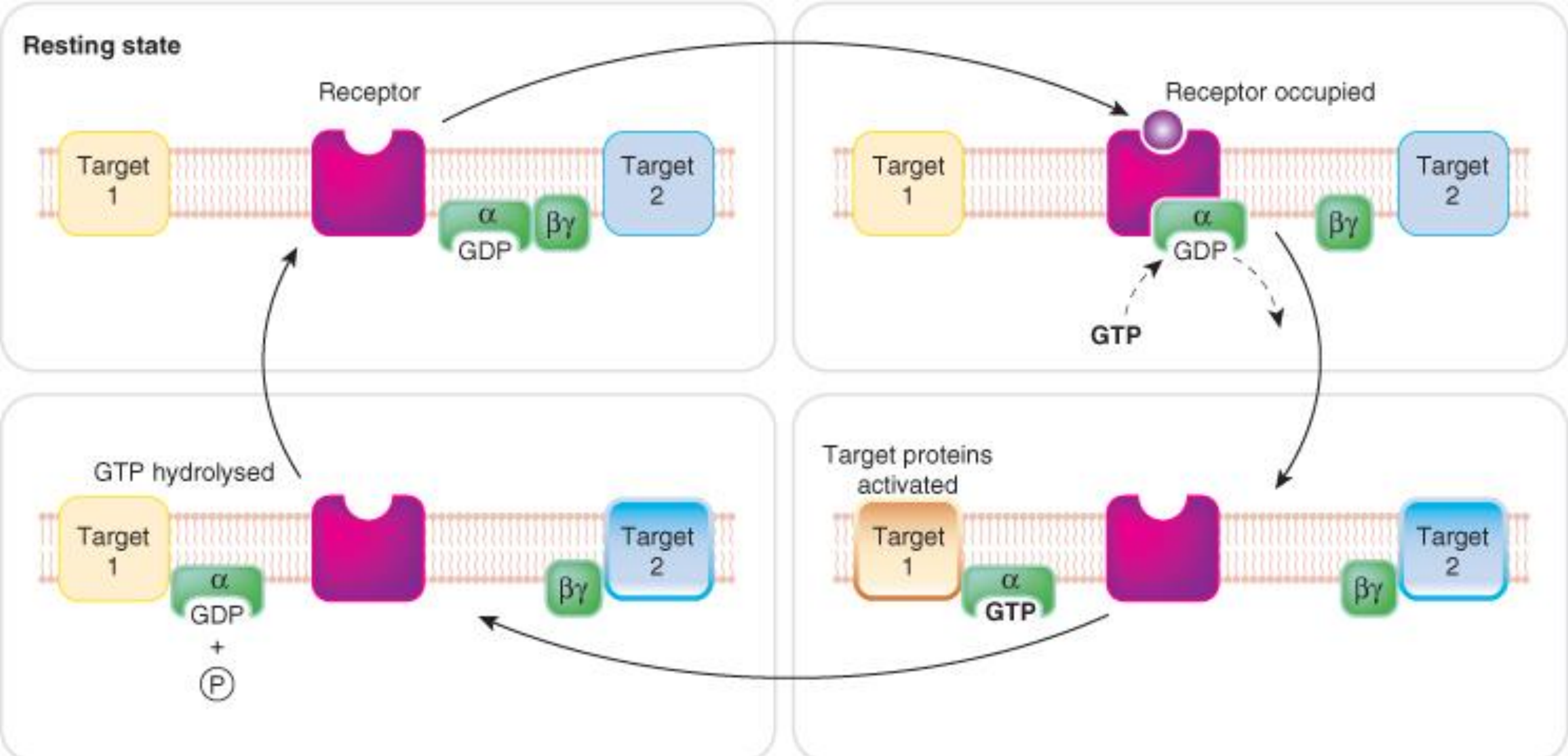
- The βγ complex gets detached from the **α** – subunit upon receptor activation.

- There are about 4 main pharmacologically important **subtypes** **(Gs, Gi, Go and Gq)**, due to molecular variation within the a subunits.

Activation of G proteins

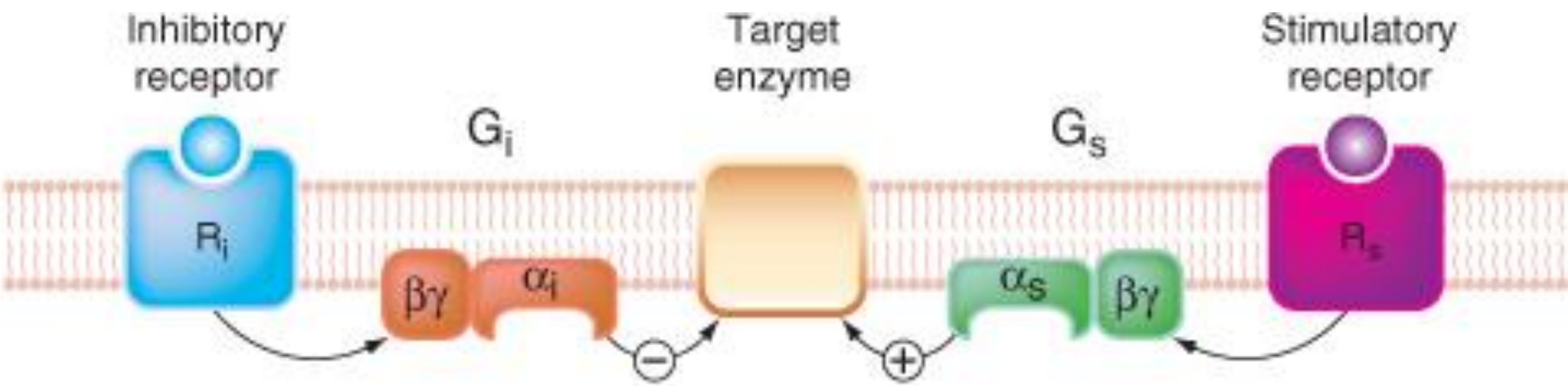


Signal Transduction: GPCRs : G proteins: subunits



Signal Transduction: GPCRs : G proteins: subunits

Example of subtype ($G\alpha_s$, $G\alpha_i$) response



What are the targets for G- proteins?

The main targets for G-proteins, through which GPCRs control different aspects of cell function are:

- **Adenylyl cyclase** : the enzyme responsible for cAMP formation

- **Phospholipase C** : the enzyme responsible for inositol phosphate and diacylglycerol (DAG) formation ion channels, particularly calcium and potassium channels

- **Rho A/Rho kinase** : a system that controls the activity of many signaling pathways controlling cell growth and proliferation, smooth muscle contraction, etc.

Targets for G- proteins cont'd

- **Ion channels:** particularly Ca^{2+} & K^{+} channels
- **Mitogen-activated protein kinase (MAP kinase):** a system that controls many cell functions, including cell division

The Main G protein subtypes and their functions

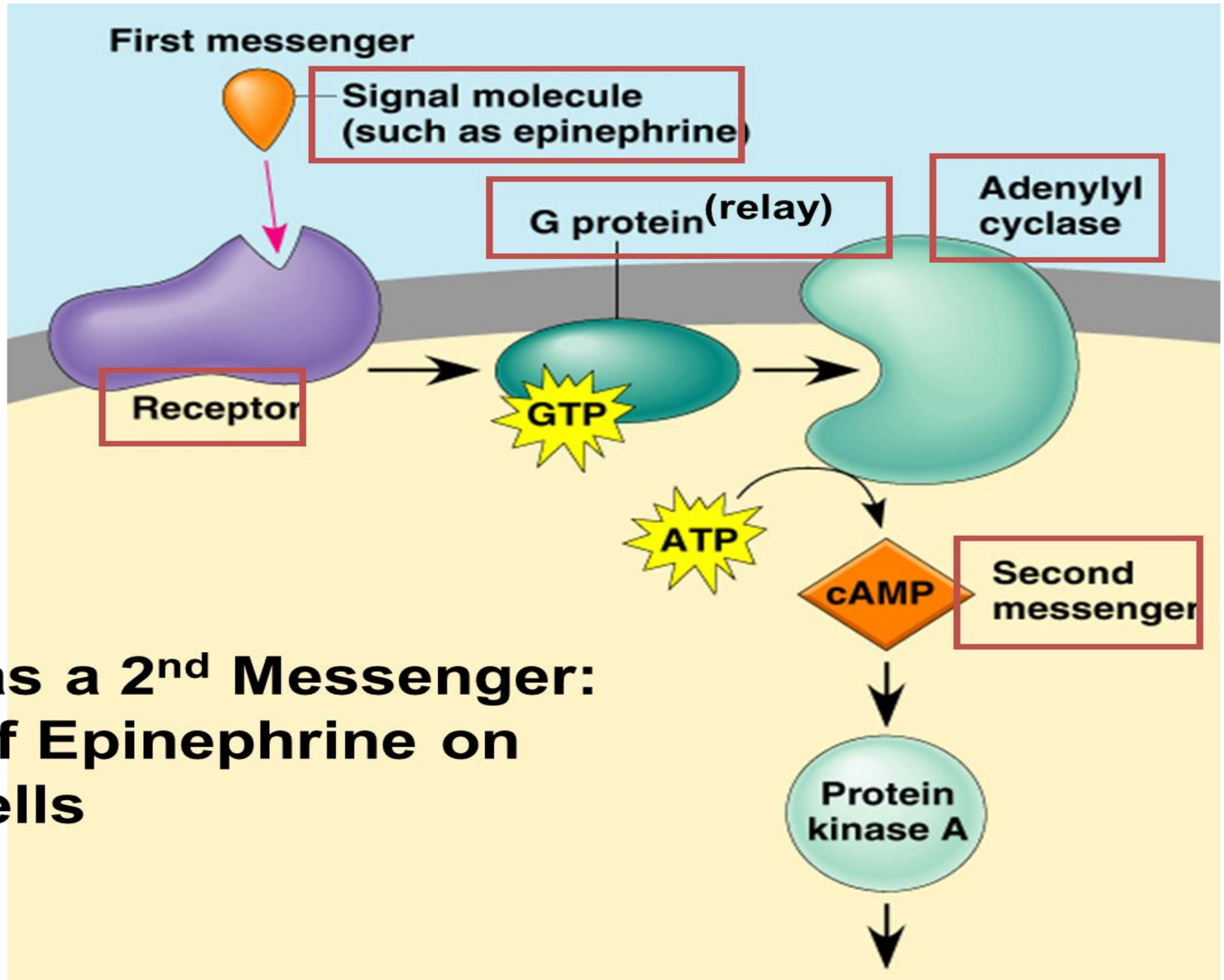
Subtype (α subunits)	Main effectors	Notes
G_s	Stimulates adenylyl cyclase, ↑ cAMP	Activated by cholera toxin, which blocks GTPase activity, thus preventing inactivation
G_i	Inhibits adenylyl cyclase, ↓cAMP	Blocked by pertussis toxin, which prevents dissociation of αβγ complex
G_q	Activates phospholipase C, ↑ inositol trisphosphate & diacylglycerol	
G_o	Effects mainly due to bg subunits	Blocked by pertussis toxin.
> 20 subtypes have been identified!		

Involvement of Second Messenger Systems

Second (2nd) Messengers

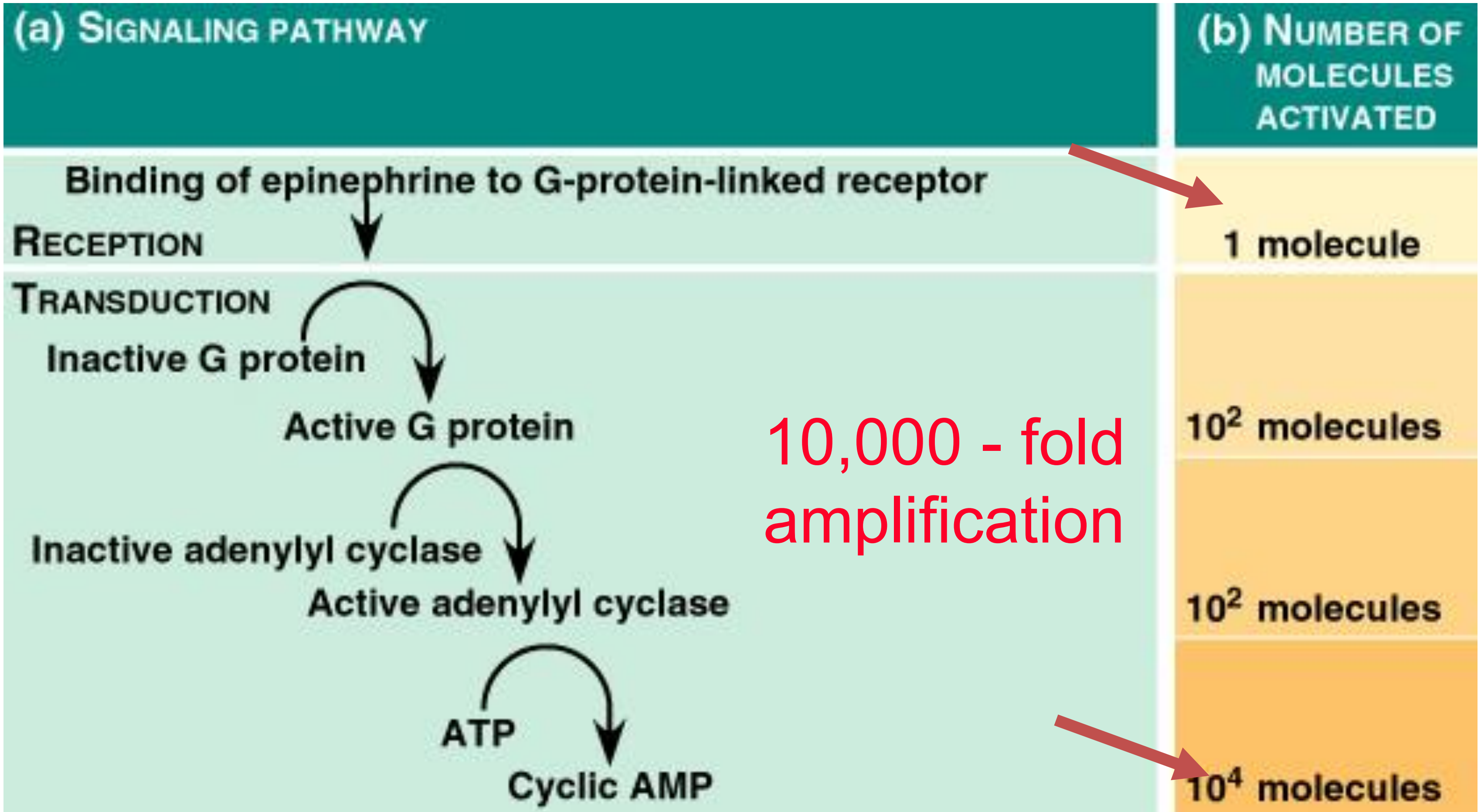
- Chemicals whose intracellular concentration increases or more rarely, decreases in response to receptor activation by agonist and which trigger processes that eventually result in a cellular response.
- **2nd messengers** greatly amplify the first message (delivered by ligand)
- The most studied 2nd messengers are:
  - Ca^{2+} ions
  - Cyclic adenosine monophosphate (cAMP)
  - Inositol-1,4,5-trisphosphate (IP3)
  - Diacylglycerol (DG)

cAMP - mediated response : Adrenaline/Epinephrine

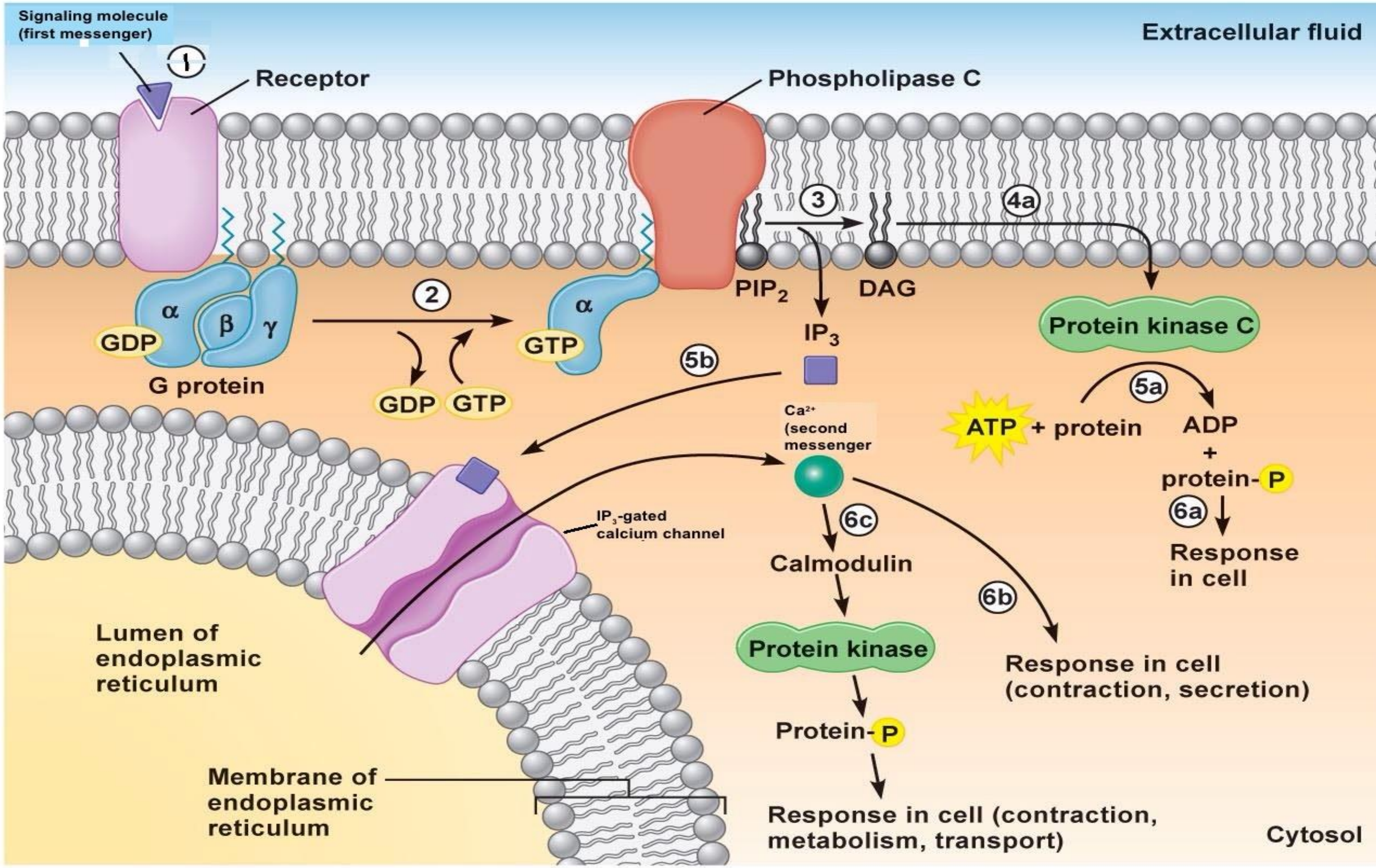


cAMP as a 2nd Messenger: Effect of Epinephrine on Liver Cells

Amplification by 2nd Messengers



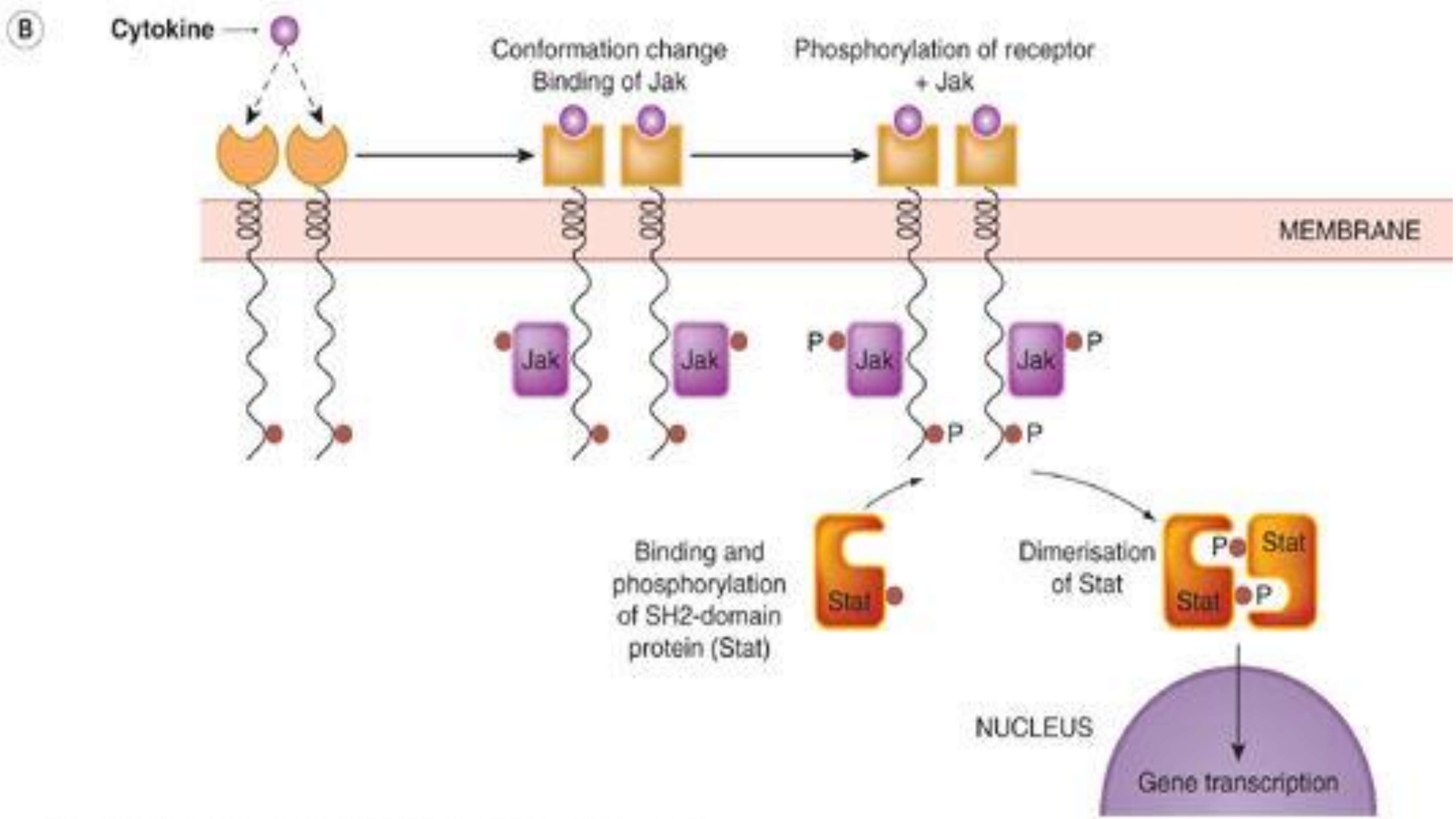
Signal Transduction Pathways



Signal Transduction Pathways

EFFECTORS	SECOND MESSENGER
Adenylate Cyclase (AC)	cAMP
Guanylate Cyclase (GC)	cGMP
Phospholipase C (PLC)	DAG and IP3
Phospholipase A (PLA_2)	Arachidonic acid
Nitric oxide Synthase	NO and CO
Ions	Na^+, Ca^{2+}, K^+, Cl^-

Kinase-linked Receptors: Cytokines

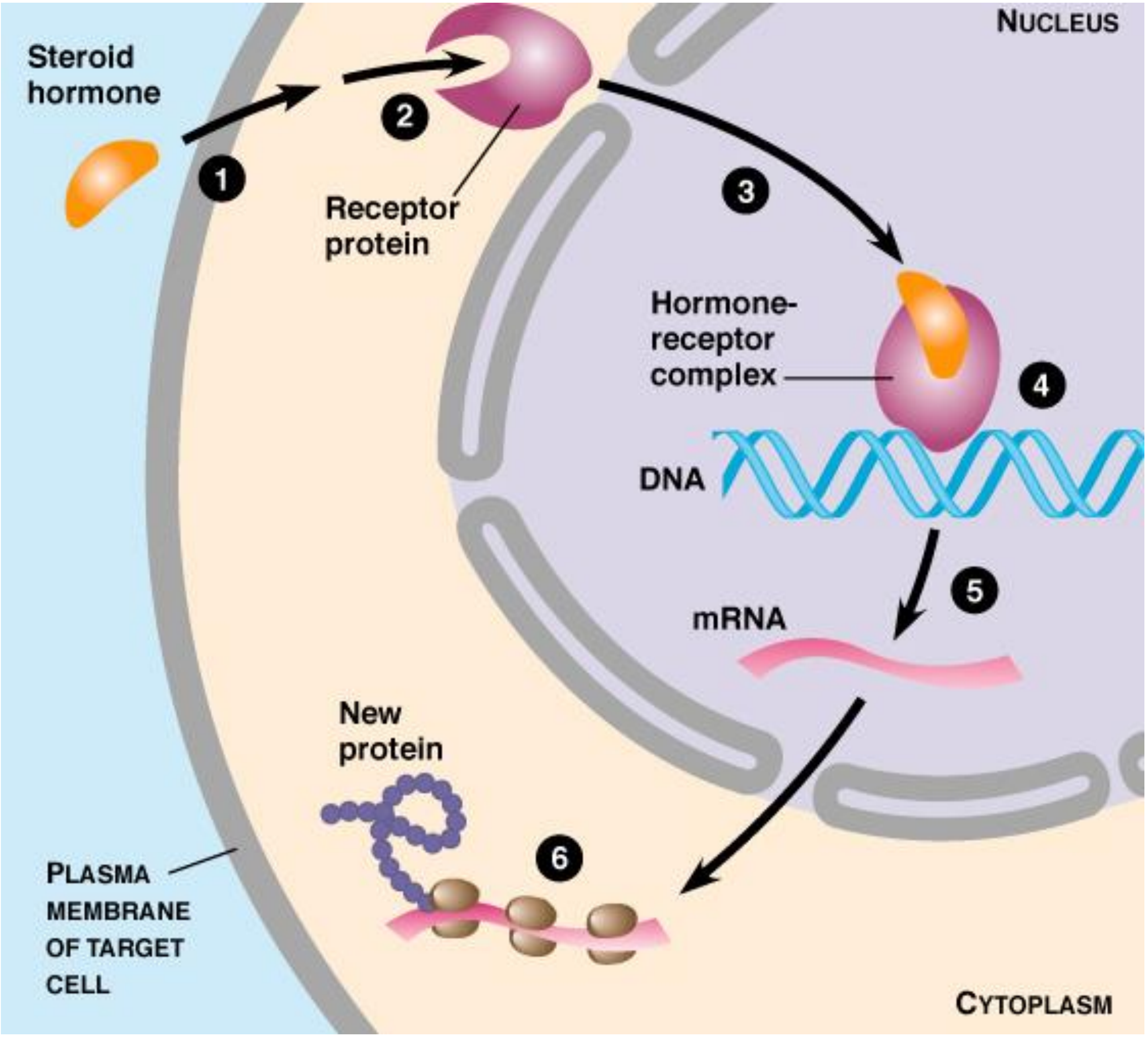


Rang et al: Rang & Dale's Pharmacology, 7e

Signal Transduction:

Nuclear Receptors

Signal transduction : Nuclear Receptor



Nuclear receptors

Examples

- Estrogen receptor
- Testosterone receptor
- Cortisol receptor
- Thyroid hormone receptors
- Aldosterone
- Peroxisome proliferator-activated receptor gamma (PPARγ)

Signalling and disease

A number of disease states are directly linked to receptor malfunction

1. Autoantibodies directed against receptor proteins
   - *myasthenia gravis* a disease of the neuromuscular junction is due to autoantibodies that inactivate nicotinic acetylcholine receptors.

2. Receptor mutations can result in activation of effector mechanisms in the absence of agonists.
   - Mutations in the receptor for thyrotropin/thyroid stimulating hormone (TSH) producing continuous over-secretion of thyroid hormone.

Signalling and disease

3. Cancer

- **Many cancers** are associated with mutations of the genes encoding growth factor receptors, kinases and other proteins involved in signal transduction

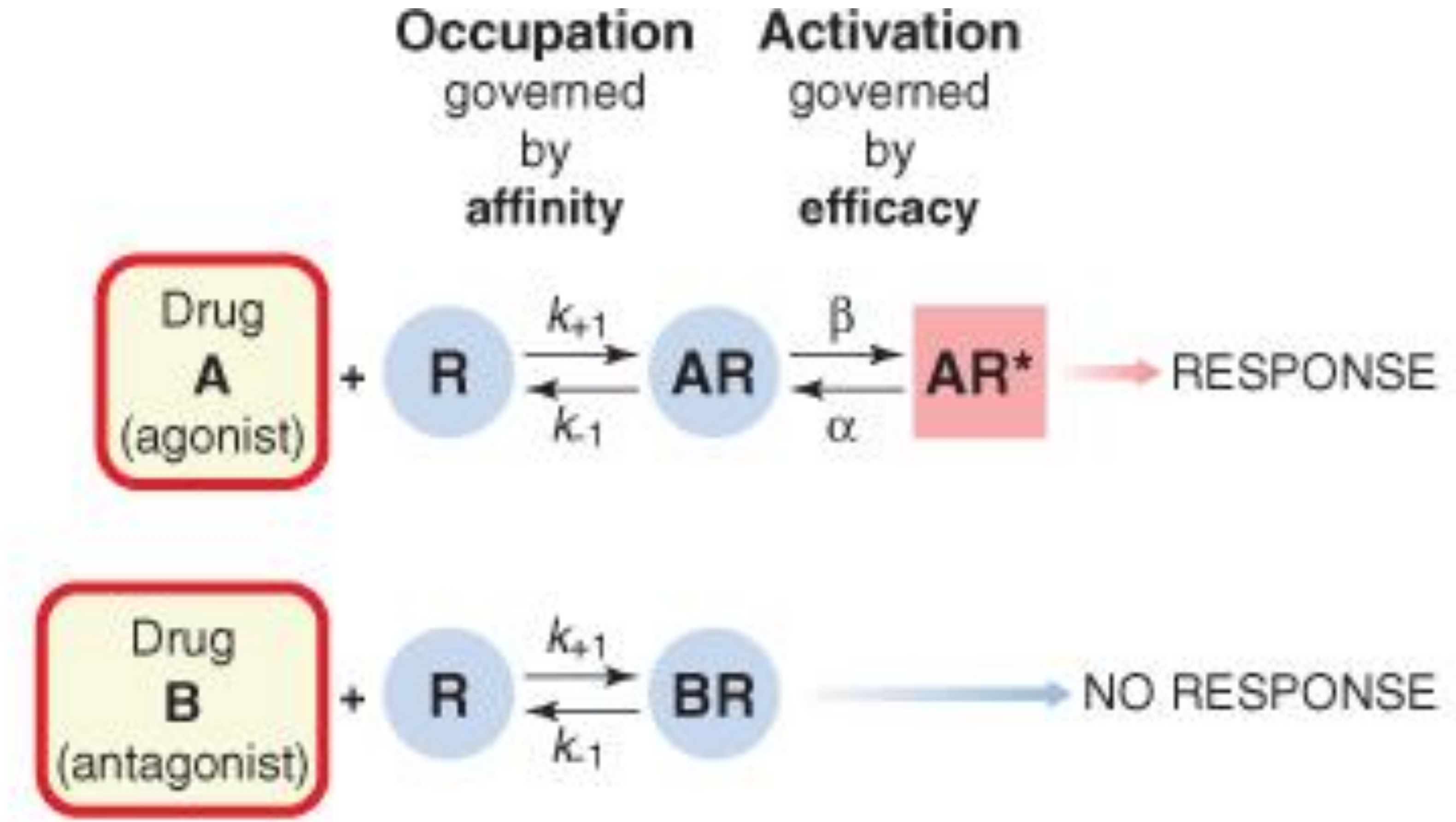
QUANTITATIVE DRUG RECEPTOR INTERACTIONS

Drug-Receptor Interactions

- Occupation of a receptor by a drug molecule may or may not result in *activation* of the receptor.

- *Activation* means that the receptor is affected by the bound molecule in such a way as to alter the receptor's behaviour towards the cell and elicit a tissue response.

Drug-Receptor Interactions; Binding and activation



Concentration-Response Relationship

Drug effect is proportional to the number of occupied receptors.

Drug (D) + Receptor (R) $\Leftrightarrow$ DR $\rightarrow$ Effect

Drugs that bind to these receptors may behave as an agonist, antagonists, partial agonist/ partial antagonist or inverse agonist

Dose-Response Relationships

- *The concentration of drug at the site of action controls the effect*
  - *Regardless of how the drug effect occurs.*

- Dose-response data are normally represented on a graph with
  - *Dose or dose fxn (log dose) on x-axis*
  - *Measured effect on y-axis*

Hypothetical dose-response curve

Features

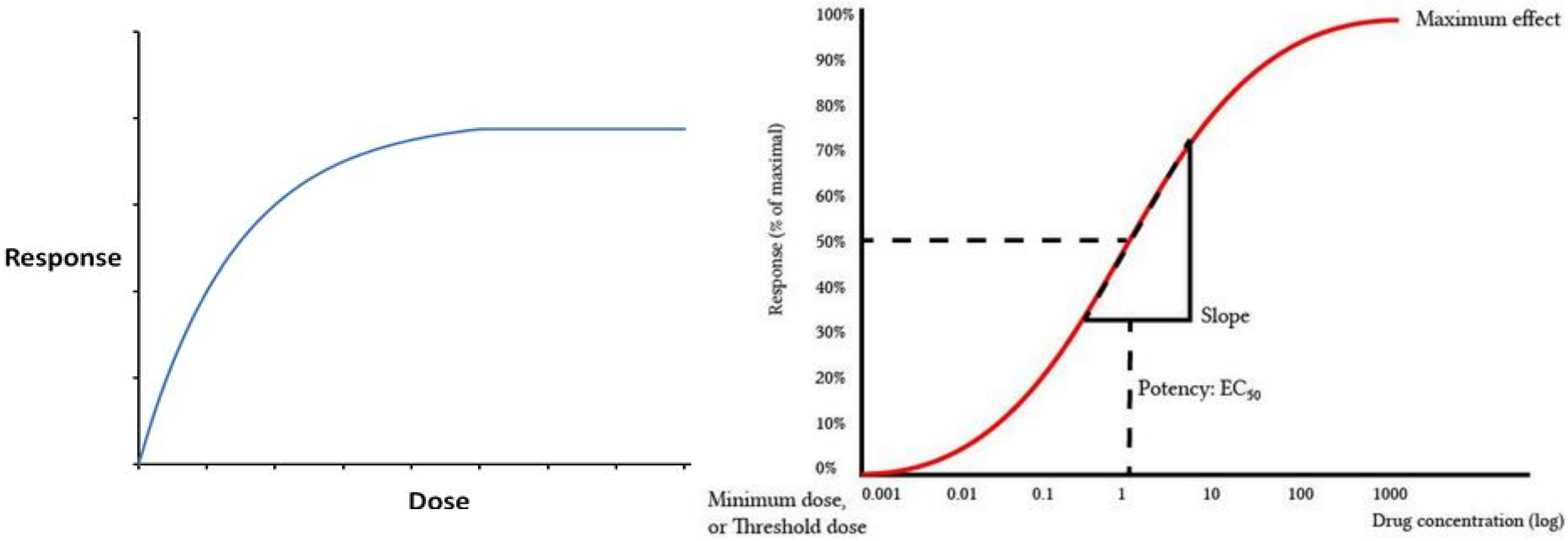
- Potency (displacement of the curve along the x-axis) - EC_{50}, pD_2, pA_2

- Maximal efficacy (greatest attainable response)

- Slope (Δ response / unit dose)

- The DRC is hyperbola or sigmoidal

Dose-response curves



Drug characteristics

Spare receptors the maximum response of a tissue may be produced by agonists when they are bound to less than 100% of the receptors. Under these circumstances the tissue is said to possess spare receptors

Drug characteristics

- **Agonists**: Are drugs capable of binding to and activating a receptor.
- **Antagonists**: are drugs that combine with receptor, but do not activate them. Antagonists reduce the probability of the transmitter substance (or another agonist) combining with the receptor and so reduce or block its action.
- **Affinity**: The tendency of a drug to bind to the receptors
- is governed by its affinity,
- **Efficacy**: whereas the tendency for it, once bound, to activate the receptor. Agonists also possess significant efficacy, whereas antagonists, in the simplest case, have zero efficacy.

Drug characteristics

Diff. types of agonists

- Agonists are drugs capable of binding to and activating a receptor.
  - ***Full Agonists :*** give maximal response.

efficacy = 1

  - ***Partial Agonists :*** can occupy a receptor but do not give maximal response

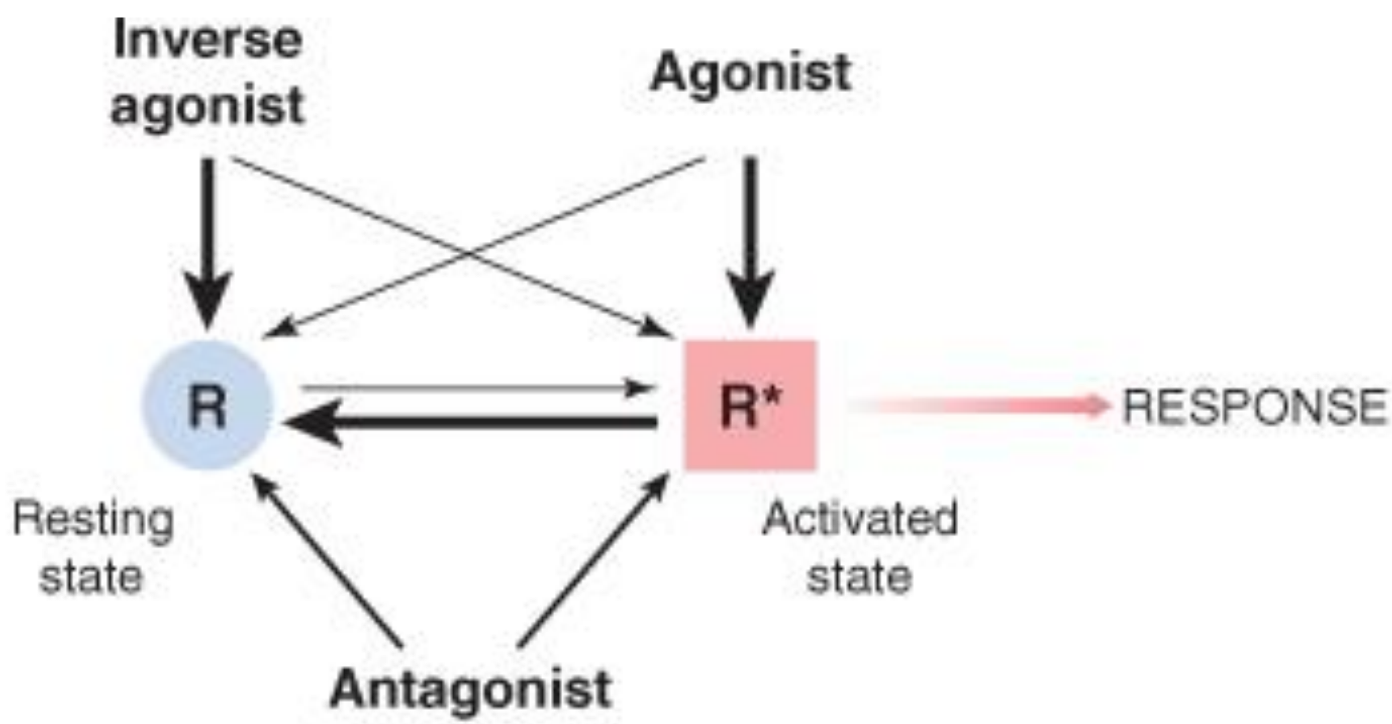
0 < efficacy <1

  - ***Inverse Agonist :*** Cause intrinsic/constitutively active targets to become inactive (negative efficacy)

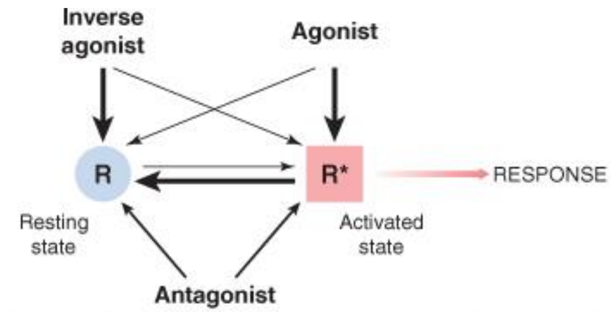
Constitutive Receptor Activity (CRA) and Inverse Agonist Effects

- Receptor proteins can spontaneously adopt an "active" conformation
  - capable of regulating cellular signaling systems in the absence of an agonist.
  - e.g delta opioid receptors NG108-15 neuroblastoma cells, constitutively activates Gi proteins in the absence of agonist

Drug characteristics

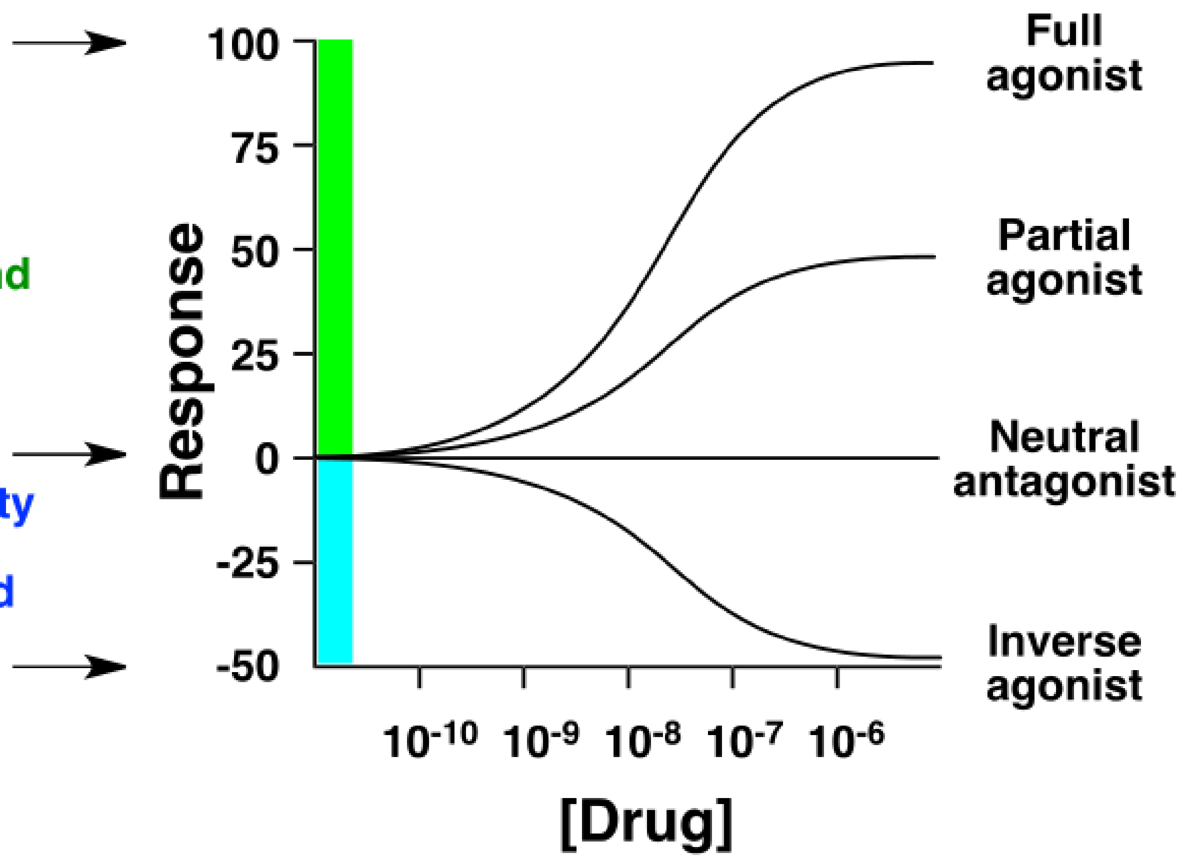


Characterization of Drug effects



activation by saturating endogenous ligand

constitutive activity of receptor in absence of ligand



Antagonism

Drug characteristics

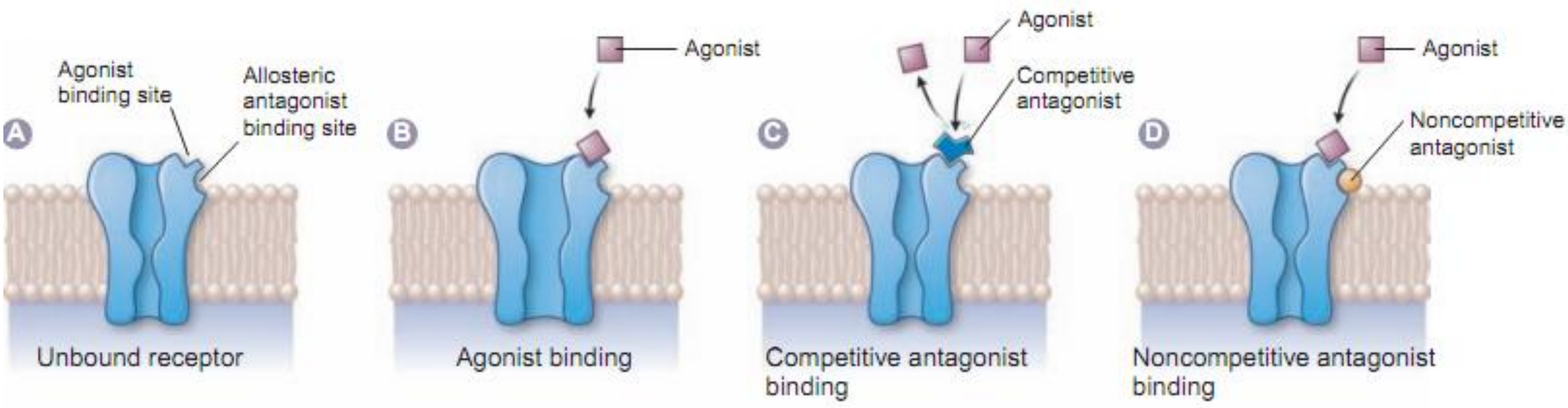
Antagonism

An antagonist is a molecule that inhibits the action of an agonist but has no effect in the absence of the agonist

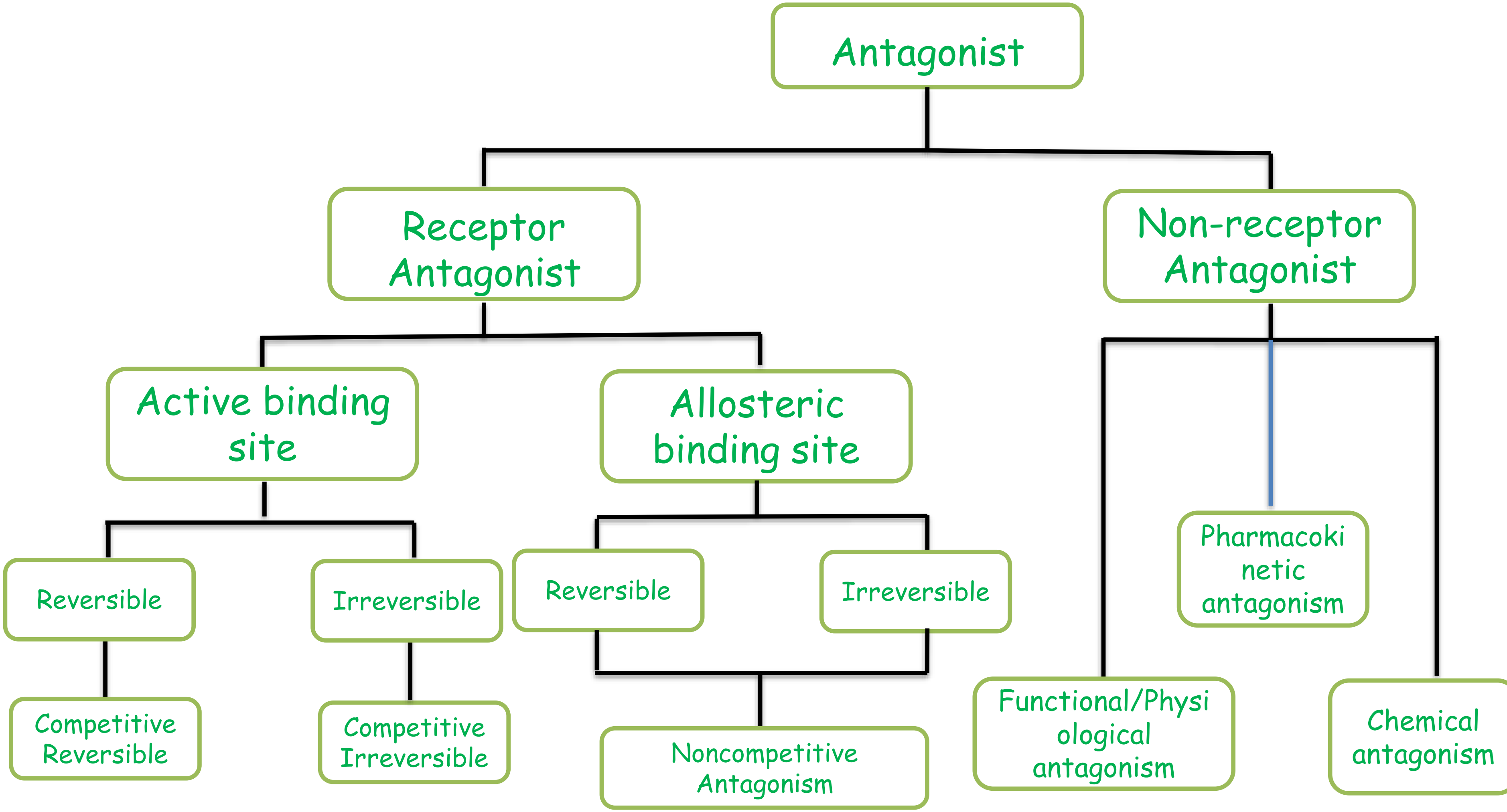
~ zero efficacy

RECEPTOR ANTAGONISM

- Mechanism of Receptor Antagonism



Types of antagonism



Antagonism

Functional/Physiological Antagonism

- Drugs produce opposing effects on the same physiological system
- **β2- adrenoceptor agonist** (eg Salbutamol) induces bronchodilation while **Cholinergic agonist** such as (methacholine) produce bronchoconstriction
- What about Adrenaline versus Acetylcholine on the heart?

Antagonism

Pharmacokinetic Antagonism

- Is the result of one drug suppressing the effect of a second drug by reducing its absorption, altering its distribution, or increasing its rate of elimination.
  - e.g. phenobarbital-induced enzyme induction increases the metabolism of the anticoagulant coumadin.

Antagonism

Chemical Antagonism

Chemical antagonism refers to the uncommon situation where the two substances combine in solution to cancel an effect.

Examples

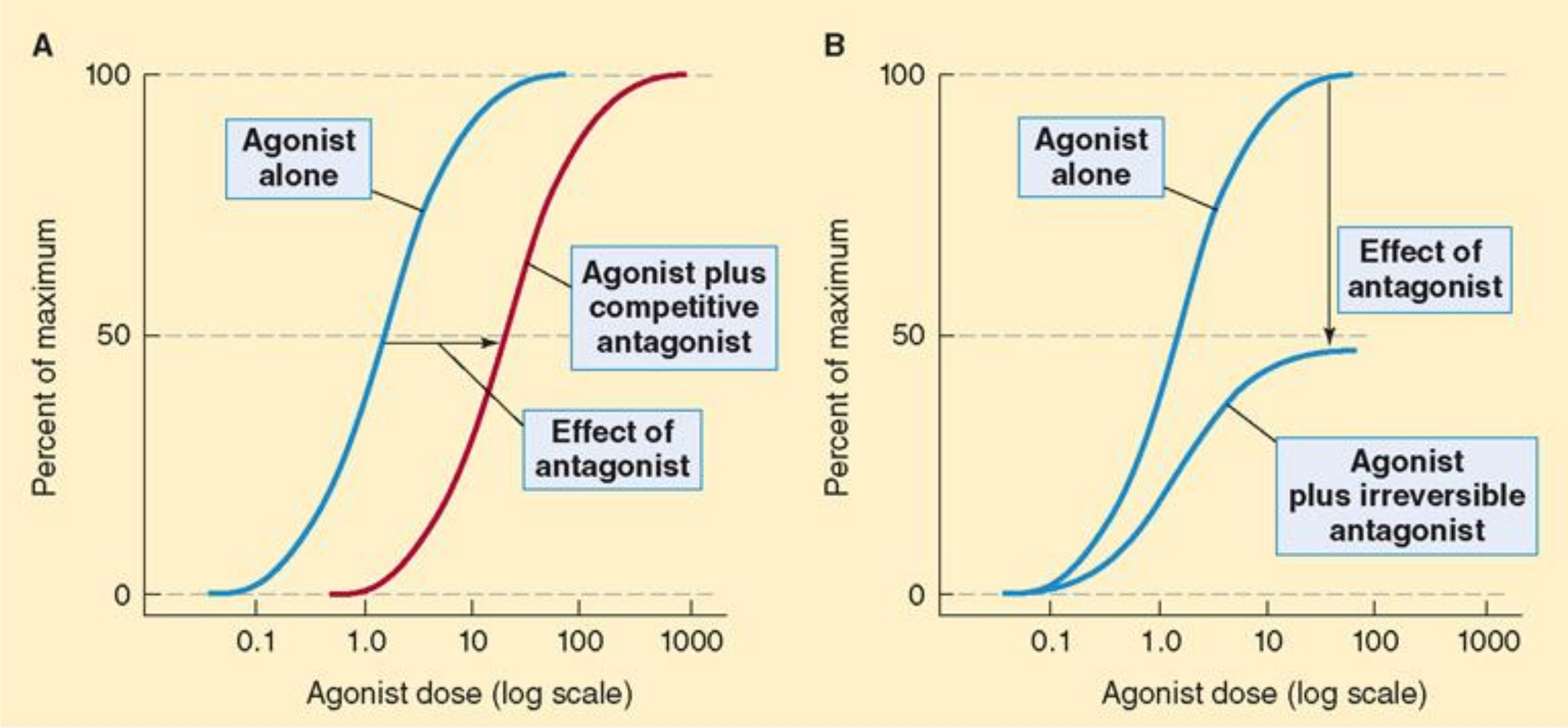
- the use of **chelating agents** (e.g. dimercaprol) that bind to heavy metals and thus reduce their toxicity
- The **neutralising antibody** (infliximab) which has an anti-inflammatory action due to its ability to sequester the inflammatory cytokine tumour necrosis factor (TNF)

MEASURING DRUG ACTION

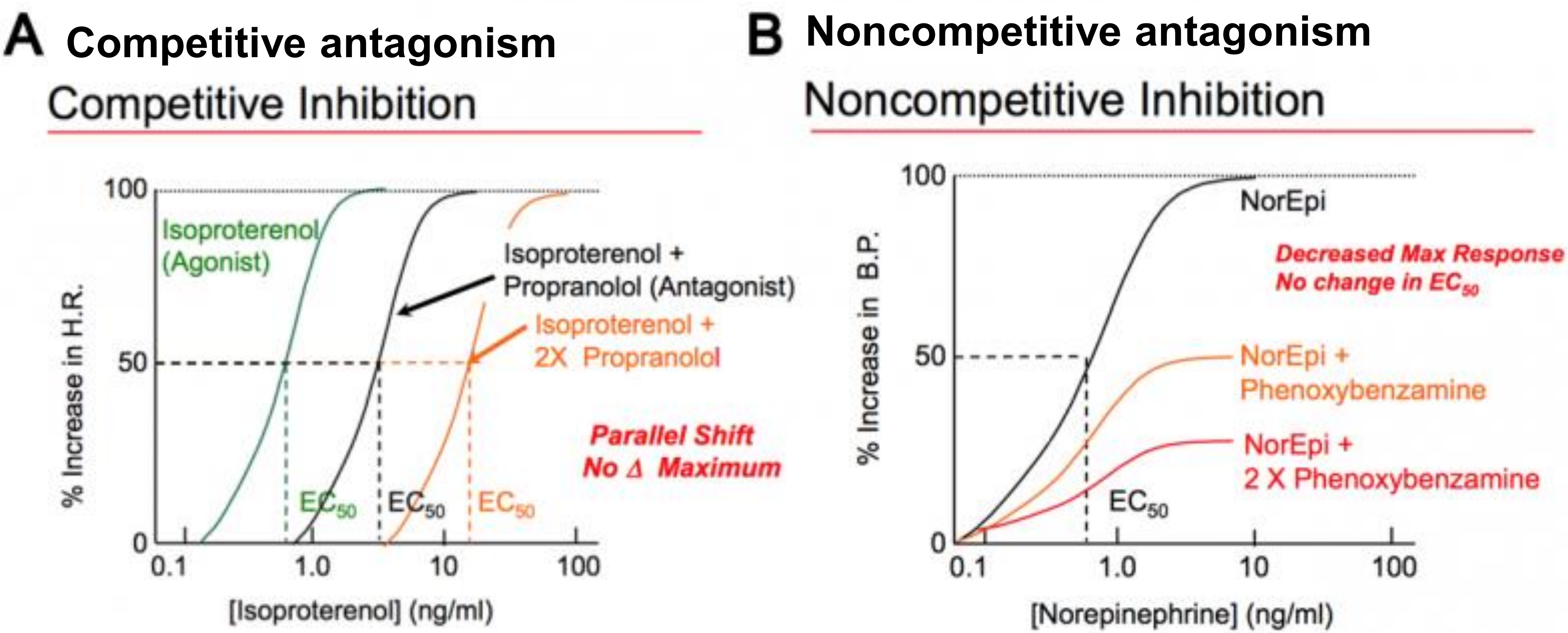
-Antagonism

Receptor Antagonism

Competitive antagonism | **Noncompetitive antagonism**



MEASURING DRUG ACTION



Drug characteristics

Drug Selectivity

- Is the degree to which a drug acts on a given **receptor** relative to other receptors.

Eg 1. Adrenomimetic agents

- $\alpha_1, \alpha_2, \beta_1, \beta_2$ — Epinephrine
- $\alpha_1, \alpha_2, \beta_1$- Norepinephrine
- α_1 - Phenylephrine; methoxamine
- α_2 - Clonidine
- β_1, β_2 - Isoprenaline (isoproterenol)
- β_1 - Dobutamine
- β_2 - Terbutaline; salbutamol, metaproterenol

Drug characteristics

DRUG SPECIFICITY

- Is the degree to which a drug acts on a given **organ/system** relative to other organs/systems

.

Eg Histamine H_1 receptor antagonist

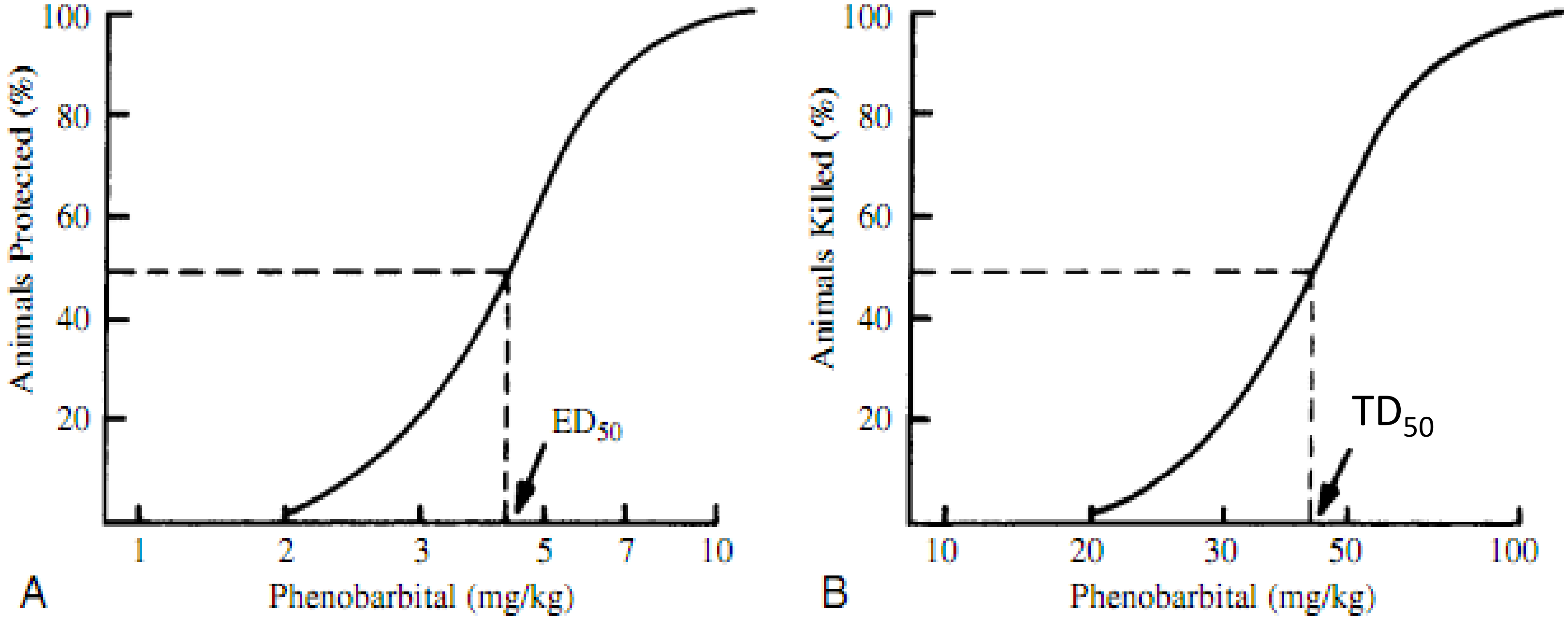
- **immune system** - Antiinflammatory
- **CNS** - sedation

MEASURING DRUG ACTION

- **ED_{50}** is the drug dose that causes a therapeutic response in 50% of the population.
- **EC_{50}** is the drug concentration that causes a 50% of the maximal response.
- **IC_{50}** is the drug concentration of an inhibitor that causes 50% inhibition.
- **TD_{50}** is the drug dose that causes toxicity in half of the study population

Drug Safety

Drug Safety


100
80
60
40
20
Animals Protected (%)
ED$_{50}$
1
2
3
5
7
10
A
Phenobarbital (mg/kg)
100
80
60
40
20
Animals Killed (%)
TD$_{50}$
10
20
30
50
100
B
Phenobarbital (mg/kg)

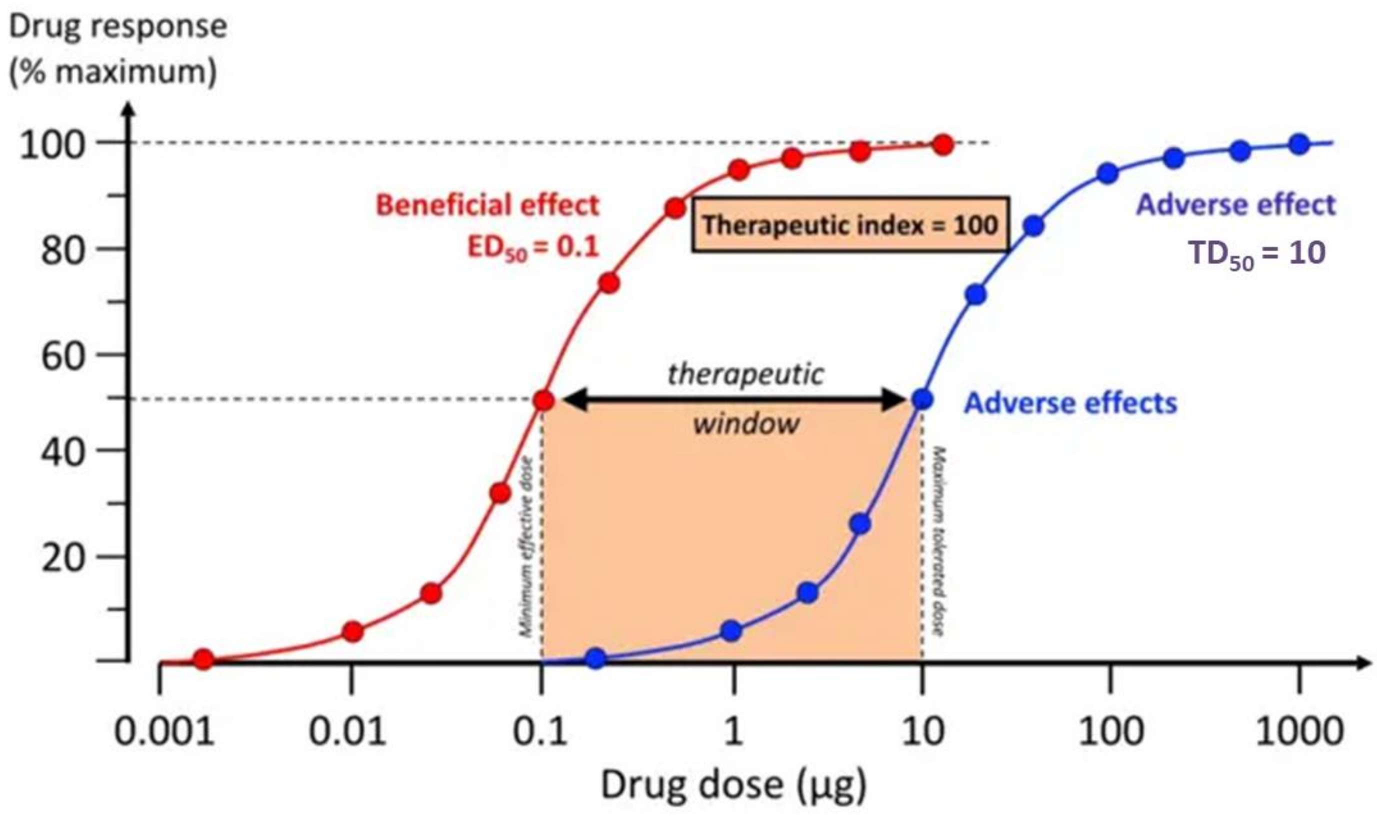
Drug Safety

Therapeutic Window

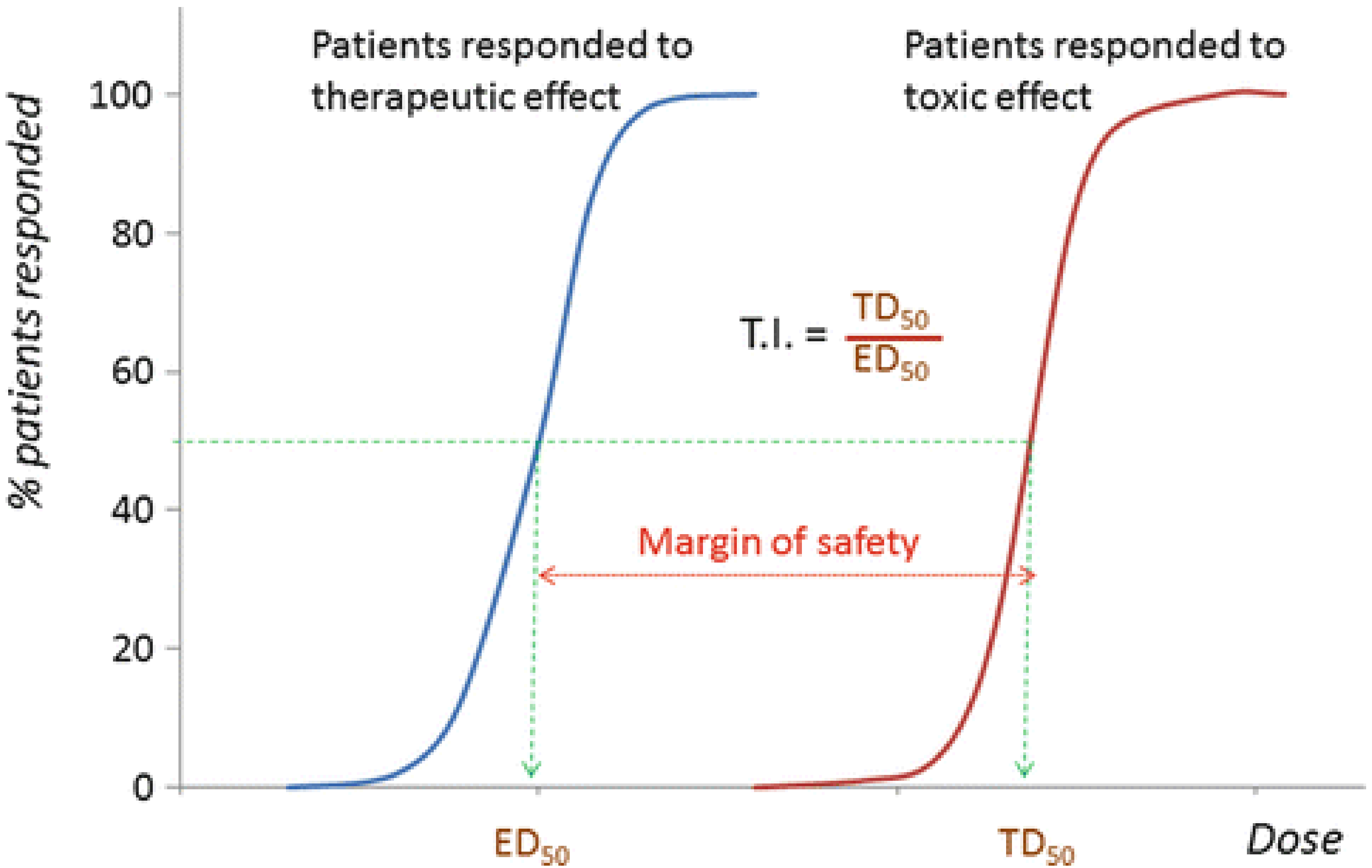
- The range of doses of a drug that elicits a therapeutic response, without unacceptable adverse effects (toxicity) in a population.

Therapeutic Index/Ratio (TI)

- Is one method of estimating a drug's margin of safety

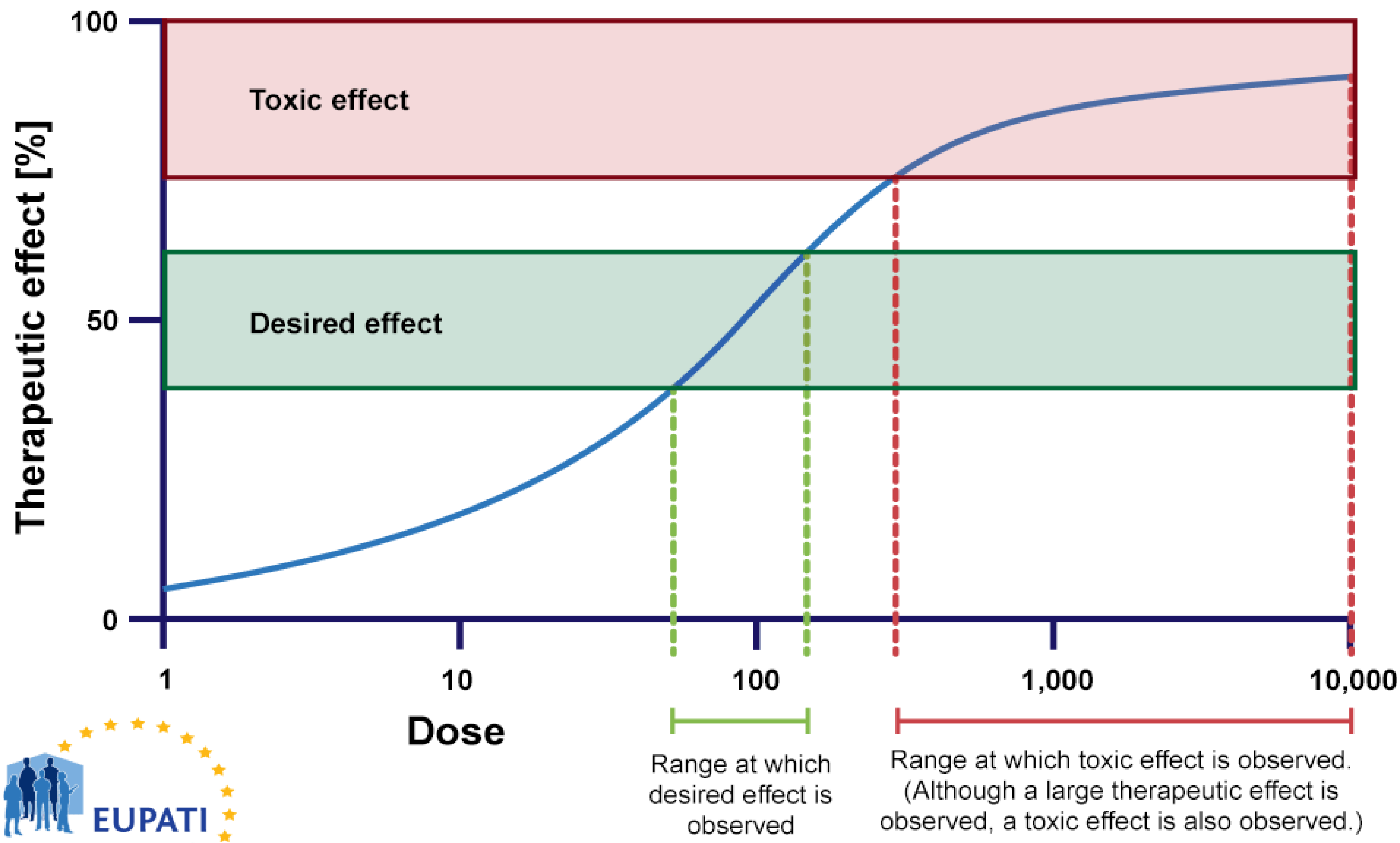


Drug Safety



Patients responded to therapeutic effect
Patients responded to toxic effect
100
80
60
40
20
0
% patients responded
$T.I. = \frac{TD_{50}}{ED_{50}}$
Margin of safety
ED_{50}
TD_{50}
Dose

Drug Safety



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DECREASED RESPONSIVENESS TO DRUGS

Desensitization/ Tachyphylaxis

- Rapid loss of drug effect (minutes)

Tolerance

- *is conventionally used to* describe a more *gradual* decrease in responsiveness to a drug, taking hours, days or weeks to develop

Refractoriness

- used mainly in relation to a loss of therapeutic efficacy

Drug resistance

- *is a term used to describe the loss* of effectiveness of antimicrobial or antitumor drugs.

Mechanisms that give rise to loss of Drug effect

- change in receptors
- translocation of receptors
- exhaustion of mediators
- increased metabolic degradation of the drug
- physiological adaptation
- active extrusion of drug from cells (mainly relevant in cancer chemotherapy)

Thank You